Image Processing (303105381)

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Lecturer

Computer science and Engineering

UNIT-3

Image filtering in the frequency domain

Outline

1. Background
2. Introduction of the Fourier Transform and the Frequency Domain
3. Smoothing Frequency- Domain Filters
4. Sharpening Frequency Domain Filters
5. Homomorphic Filtering

Outcome of this unit:

After completing this unit students will be able to:

- Understand the background of frequency domain
- To learn basic Fourier Transform and the frequency domain.
- To understand smoothing
- To understand sharpening in frequency domain.
- To get Homomorphic filtering.

1. Background

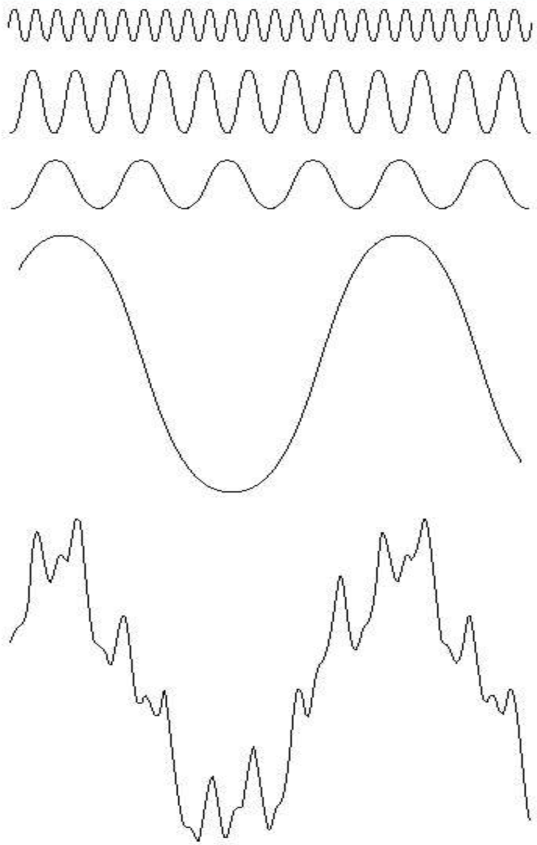


FIGURE 4.1 The function at the bottom is the sum of the four functions above it. Fourier's idea in 1807 that periodic functions could be represented as a weighted sum of sines and cosines was met with skepticism.

- The **frequency domain** refers to the plane of the two dimensional discrete Fourier transform of an image.
- The purpose of the Fourier transform is to represent a signal as a linear combination of sinusoidal signals of various frequencies.

Fourier Series and Fourier Transform: History

Jean Baptiste Joseph Fourier, French mathematician and physicist (03/21/1768-05/16/1830) http://en.wikipedia.org/wiki/Joseph_Fourier

Orphaned: at nine

Egyptian expedition with **Napoleon I**: 1798
Governor of Lower Egypt

Permanent Secretary of the French Academy of Sciences: 1822

Théorie analytique de la chaleur : 1822

(**The Analytic Theory of Heat**)

Nobody paid much attention when the work was first published.
It one of the most important mathematical theories in modern engineering

Fourier Series and Fourier Transform

- Fourier Series

  Any periodic function can be expressed as the sum of sines and /or cosines of different frequencies, each multiplied by a different coefficients

- Fourier Transform

  Any function that is not periodic can be expressed as the integral of sines and /or cosines multiplied by a weighing function

2. Introduction to the Fourier Transform and Frequency Domain

Fourier Tr. and Frequency Domain

Time, spatial Domain Signals

Fourier Tr. →

← Inv Fourier Tr.

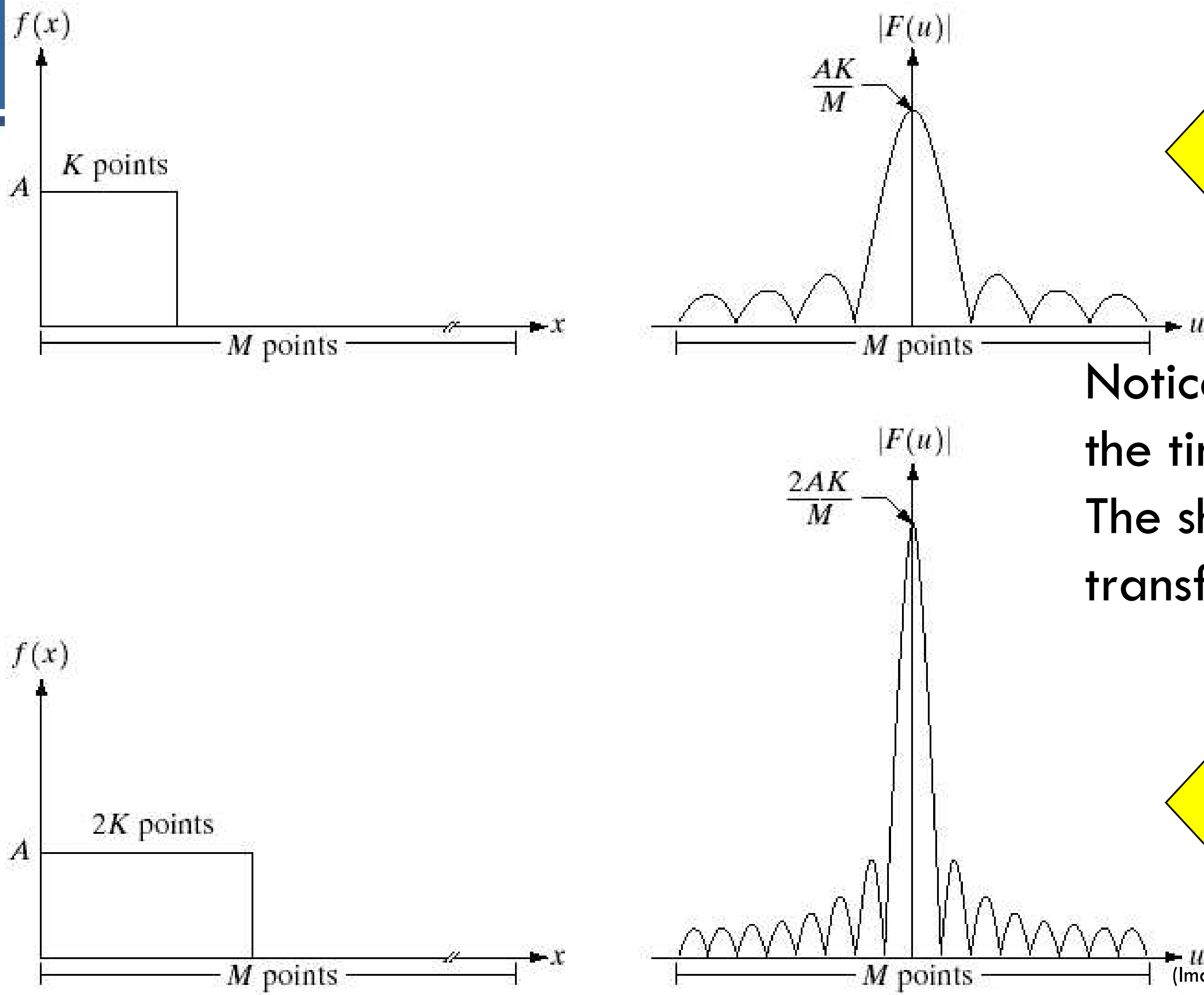
Frequency Domain Signals

1-D, Continuous case

Fourier Tr.:

Inv. Fourier Tr.:

Example of 1-D Fourier Transforms



Notice that the longer the time domain signal, The shorter its Fourier transform

(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Relation Between Δx and Δu

For a signal $f(x)$ with M points, let spatial resolution Δx be space between samples in $f(x)$ and let frequency resolution Δu be space between frequencies components in $F(u)$, we have

Example: for a signal f(x) with sampling period 0.5 sec, 100 point, we will get frequency resolution equal to

This means that in $F(u)$ we can distinguish 2 frequencies that are apart by 0.02 Hertz or more.

Fourier Transform: One Continuous Variable

Fourier Transform and Convolution

Fourier Transform Pairs

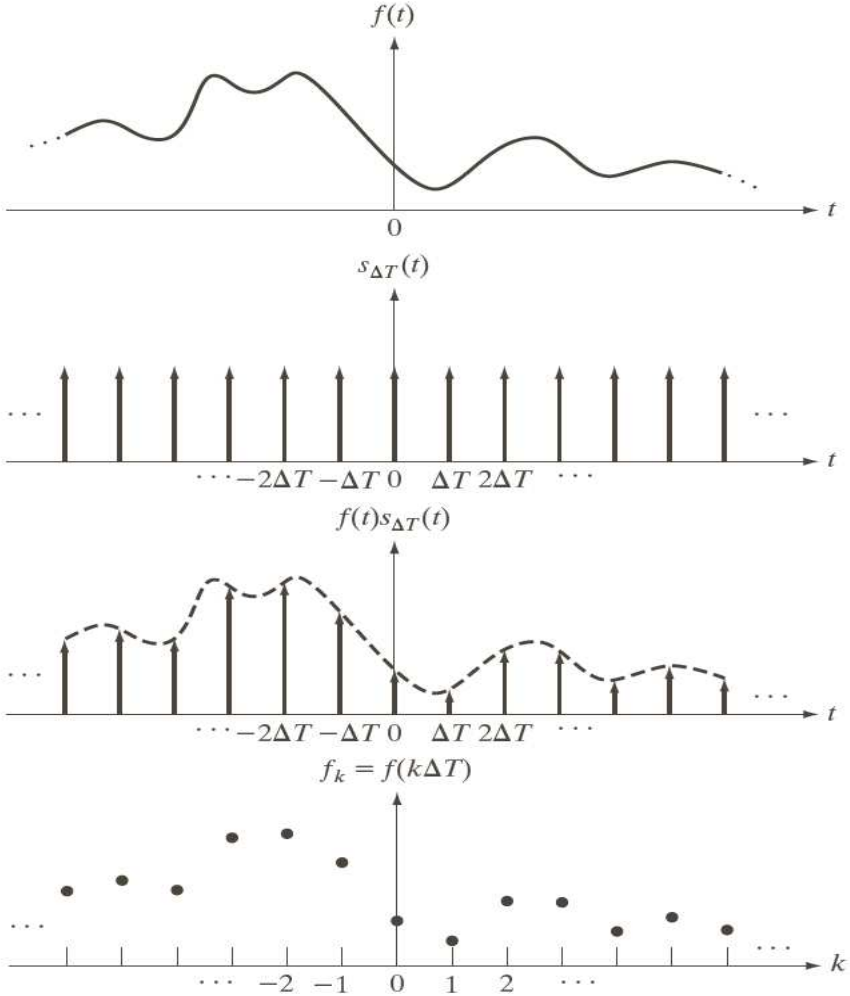
★

★

The Fourier transform of the sampled function (shown in the following figure) is

1. Continuous

2. Discrete



Introduction to the Fourier Transform and the Frequency Domain

- The one-dimensional Fourier transform and its inverse
  - Fourier transform (continuous case)
  - Inverse Fourier transform:
- The two-dimensional Fourier transform and its inverse
  - Fourier transform (continuous case)
  - Inverse Fourier transform:

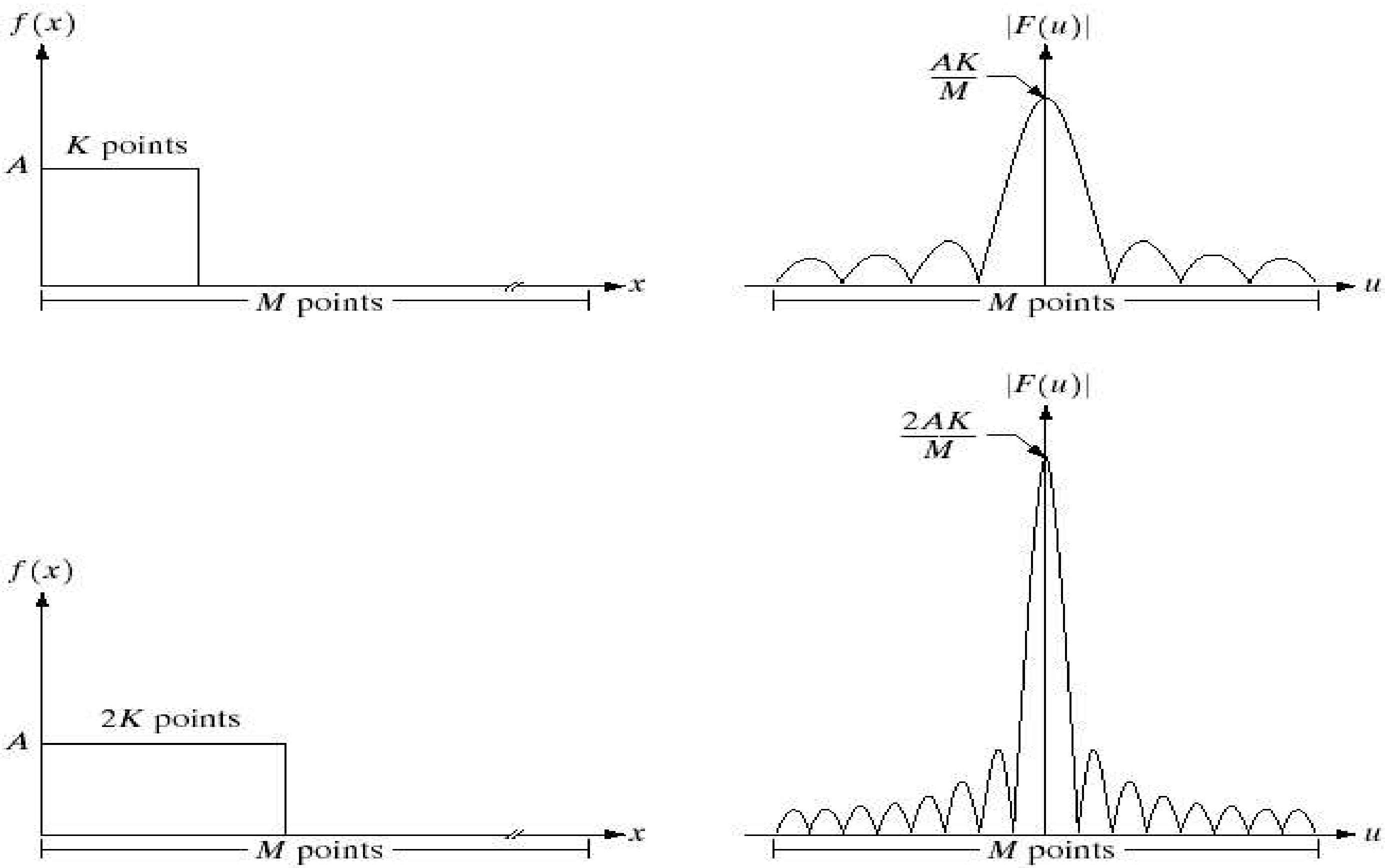
Contd…

- The one-dimensional Fourier transform and its inverse
  - Fourier transform (discrete case) DTC
  - Inverse Fourier transform:

Contd…

- $F(u)$ can be expressed in polar coordinates:
  - $R(u)$: the real part of $F(u)$
  - $I(u)$: the imaginary part of $F(u)$
- Power spectrum:

The One-Dimensional Fourier Transform Example

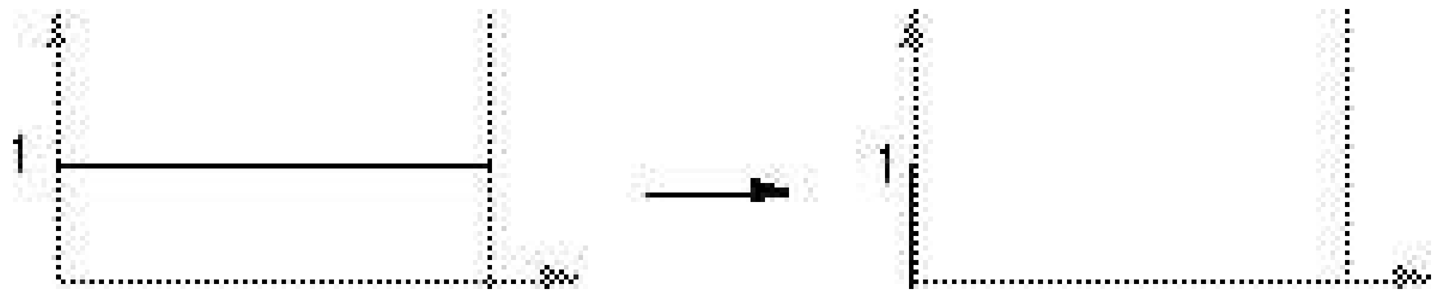


a b
c d

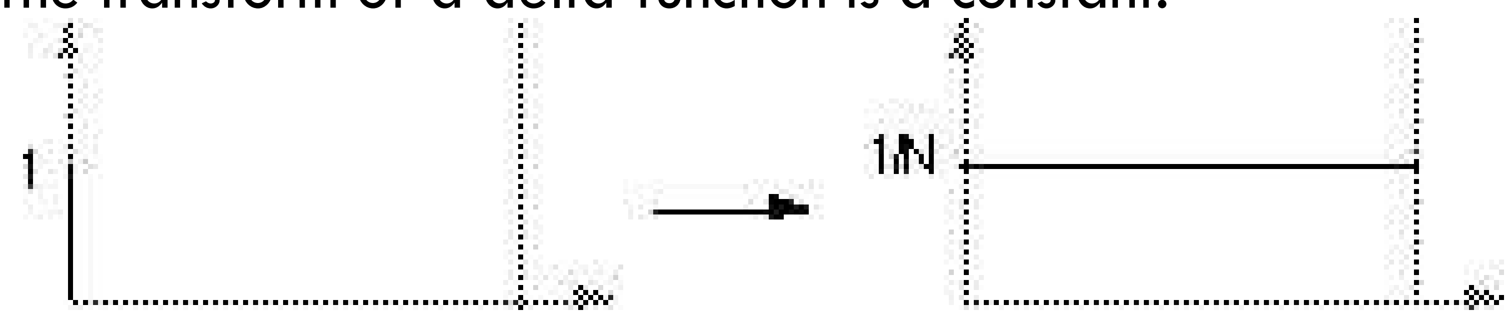
FIGURE 4.2 (a) A discrete function of M points, and (b) its Fourier spectrum. (c) A discrete function with twice the number of nonzero points, and (d) its Fourier spectrum.

The One-Dimensional Fourier Transform Example

- The transform of a constant function is a DC value only.

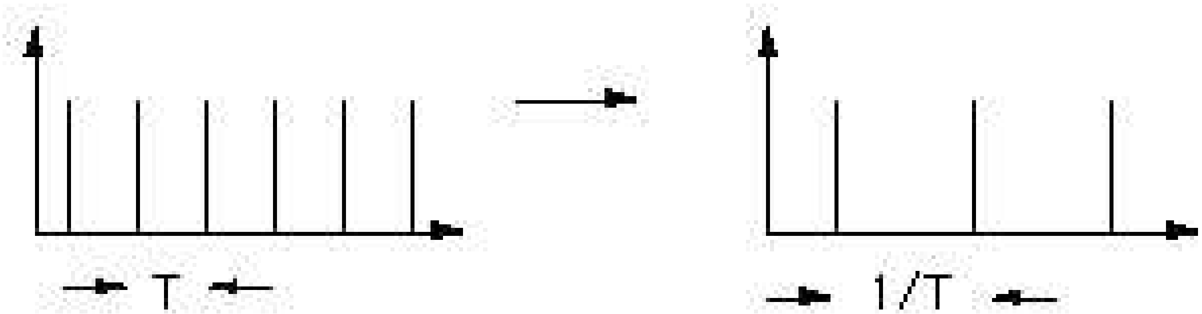


- The transform of a delta function is a constant.

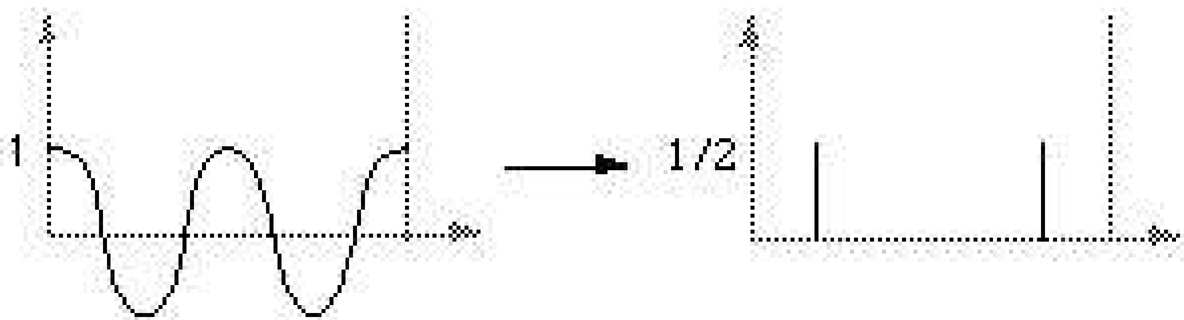


The One-Dimensional Fourier Transform Example

- The transform of an infinite train of delta functions spaced by T is an infinite train of delta functions spaced by 1/T.

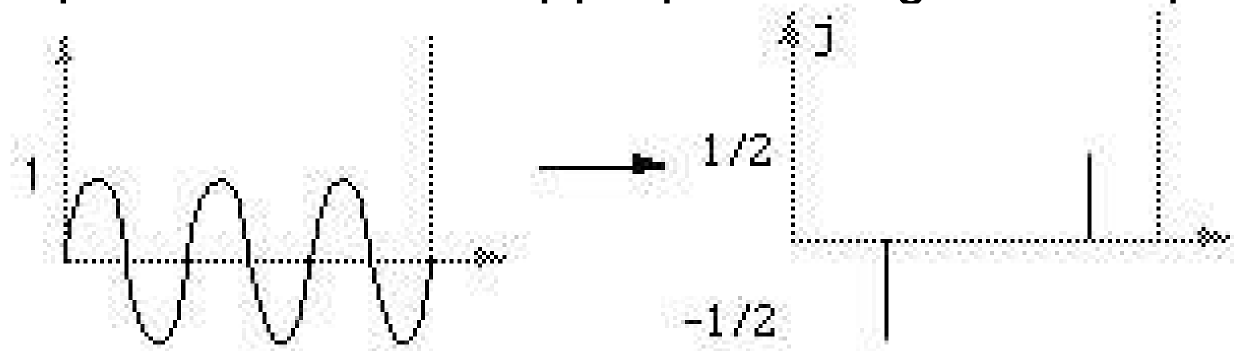


- The transform of a cosine function is a positive delta at the app

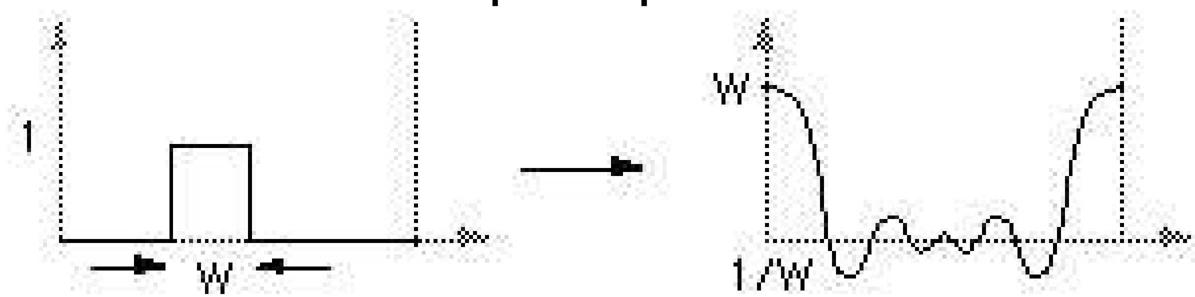


The One-Dimensional Fourier Transform Example

- The transform of a sin function is a negative complex delta function at the appropriate positive frequency and a positive complex delta at the appropriate negative frequency.



- The transform of a square pulse is a sinc function.



2-D Fourier Transform: Continuous

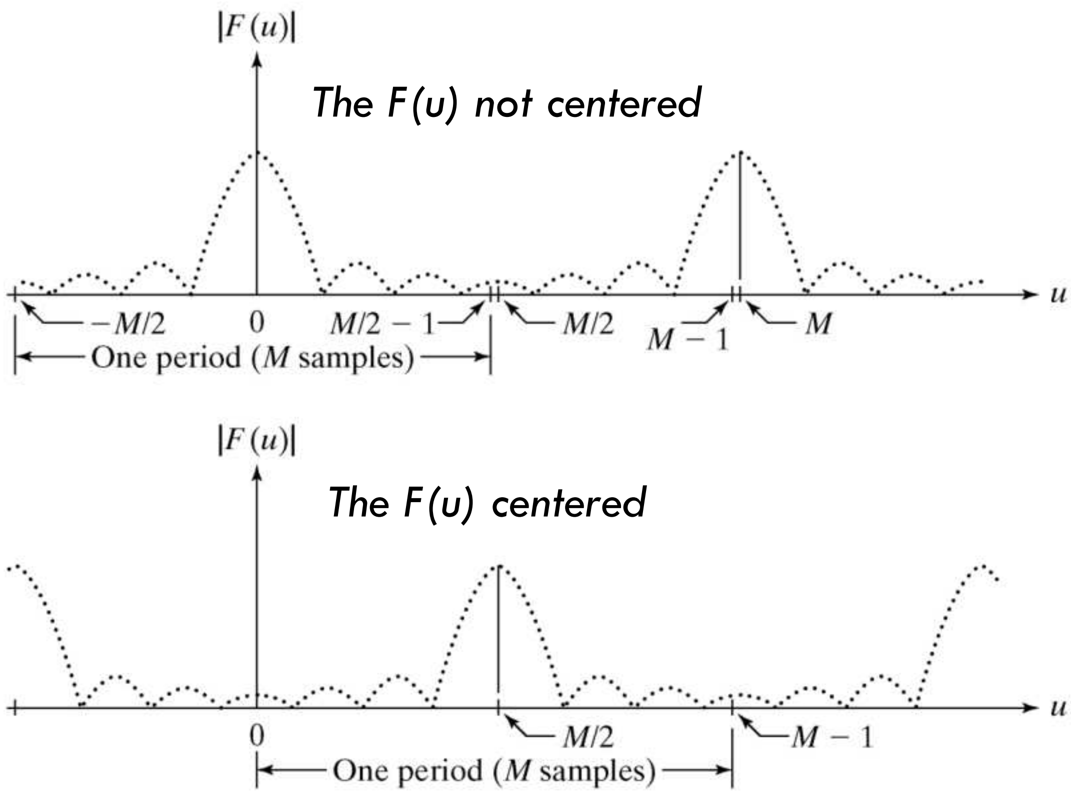
Fourier Transform and Inverse Fourier transform

Properties of the 2-D DFT

relationships between spatial and frequency intervals

Periodicity/Symmetry

The Periodicity property: F(u) in 1D DFT has a period of M
The Symmetry property: The magnitude of the transform is centered on the origin.



a
b

FIGURE 4.1 (a) Fourier spectrum showing back-to-back half periods in the interval $[0, M-1]$. (b) Centered spectrum in the same interval, obtained by multiplying $f(x)$ by $(-1)^x$ prior to computing the Fourier transform.

Periodicity

The Periodicity property: F(u,v) in 2D DFT has a period of N in horizontal and M in vertical directions

The Symmetry property: The magnitude of the transform is centered on the origin.

- Remember that we have the following in two dimensions:
- $f(x + M, y + N) = f(x, y)$
- $F(u + M, v + N) = F(u, v)$
- This implies that we only need to know $f(x, y)$ or $F(u, v)$ for one period

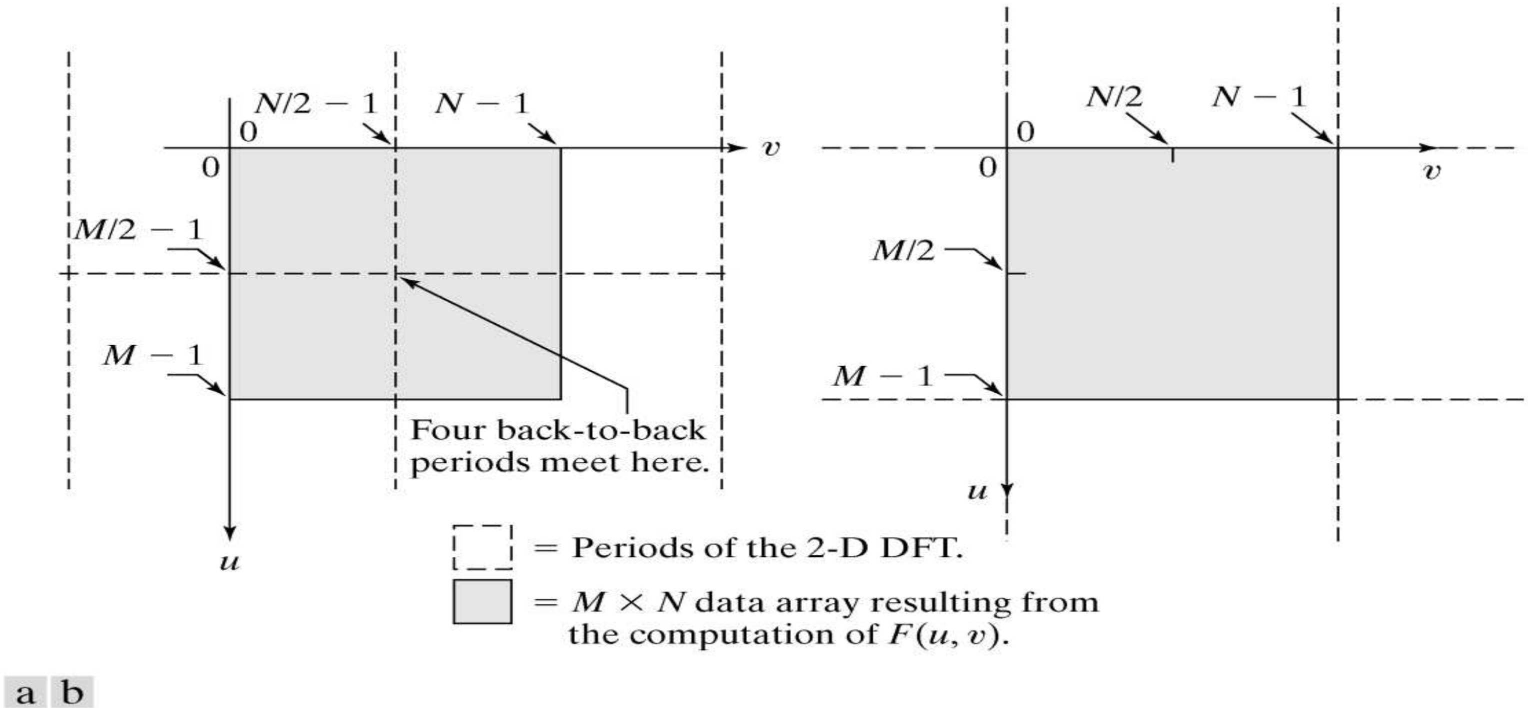


FIGURE 4.2 (a) $M \times N$ Fourier spectrum (shaded), showing four back-to-back quarter periods contained in the spectrum data. (b) Spectrum obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ prior to computing the Fourier transform. Only one period is shown shaded because this is the data that would be obtained by an implementation of the equation for $F(u, v)$.

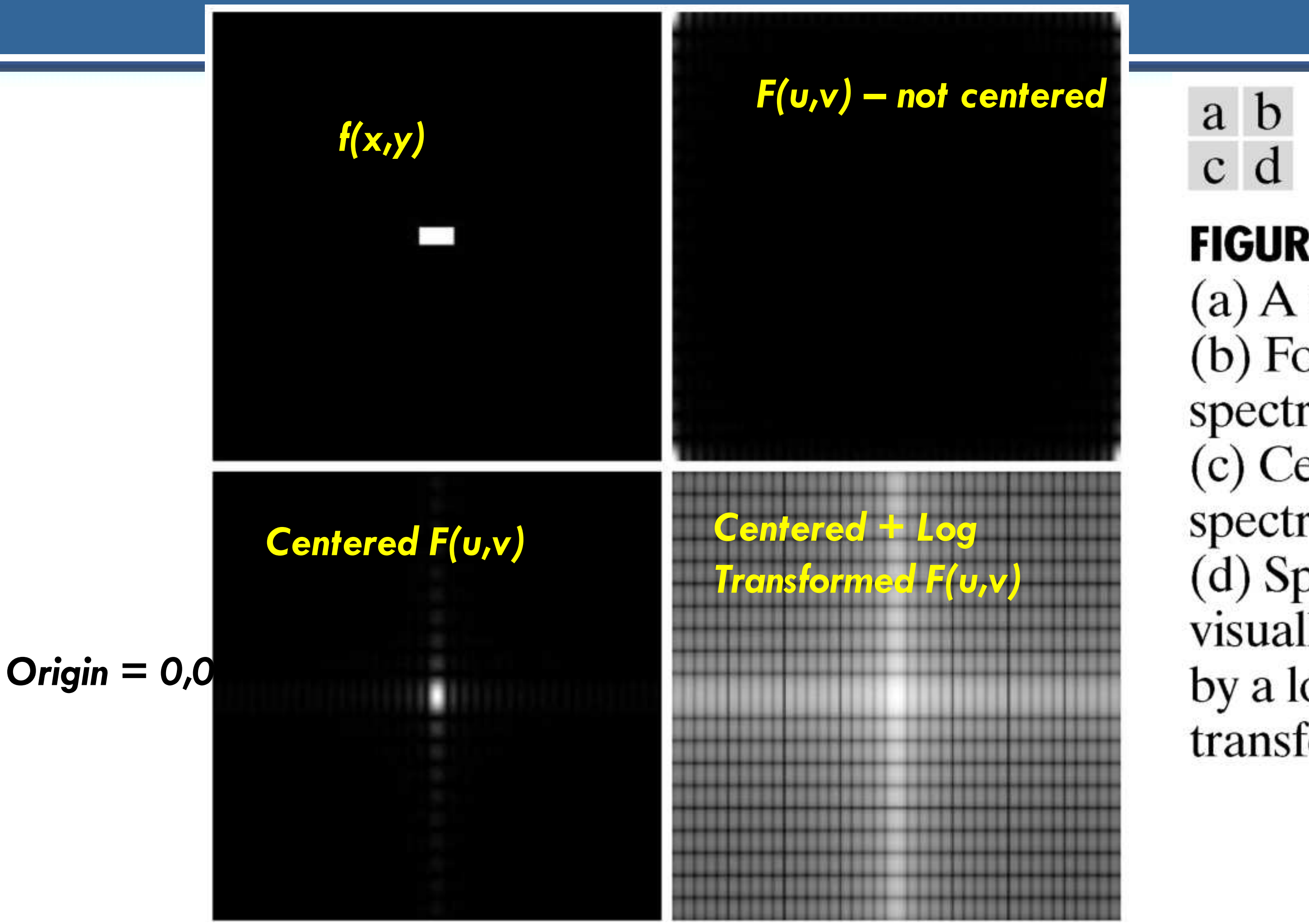
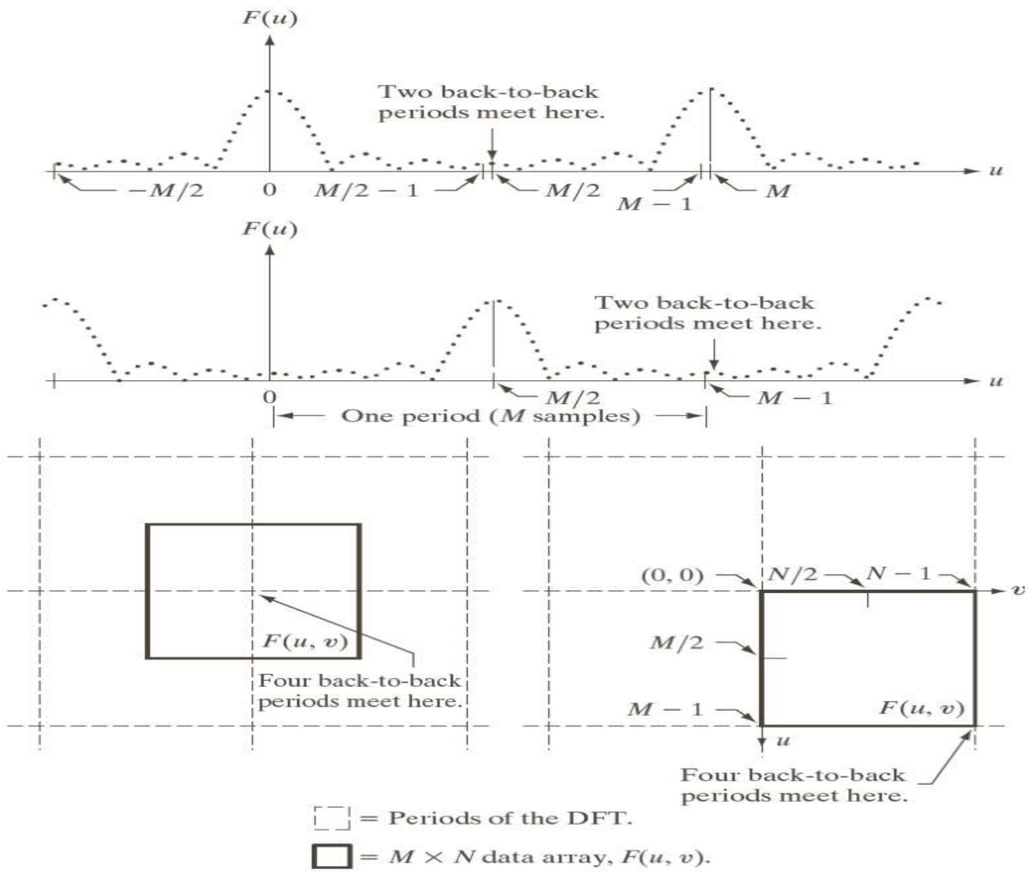


FIGURE 4.3
(a) A simple image.
(b) Fourier spectrum.
(c) Centered spectrum.
(d) Spectrum visually enhanced by a log transformation.

Properties of the 2-D DFT

periodicity

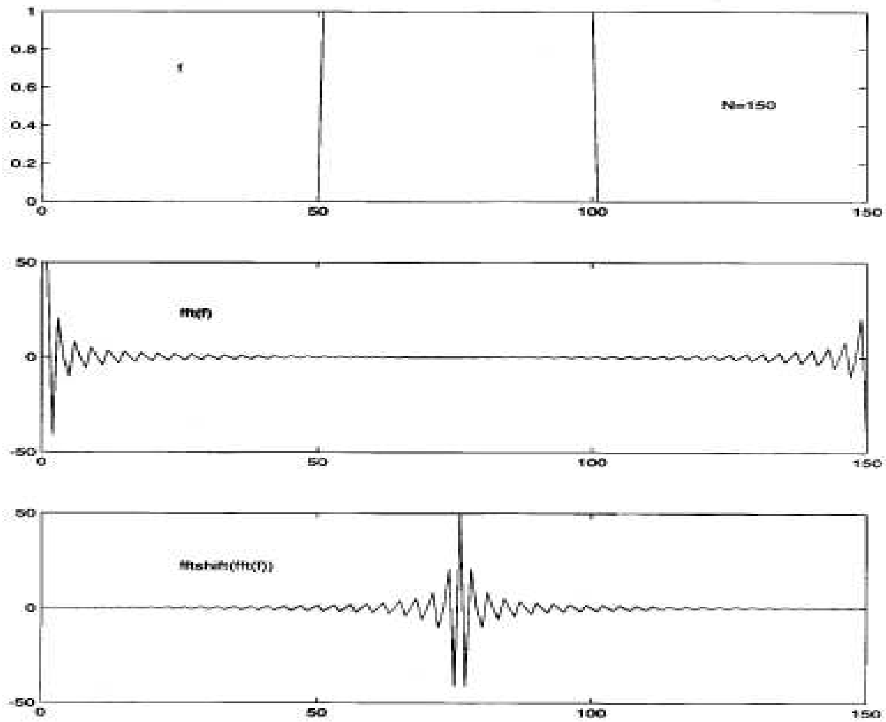


a
b
c d

FIGURE 4.23 Centering the Fourier transform. (a) A 1-D DFT showing an infinite number of periods. (b) Shifted DFT obtained by multiplying $f(x)$ by $(-1)^x$ before computing $F(u)$. (c) A 2-D DFT showing an infinite number of periods. The solid area is the $M \times N$ data array, $F(u, v)$, obtained with Eq. (4.5-15). This array consists of four quarter periods. (d) A Shifted DFT obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ before computing $F(u, v)$. The data now contains one complete, centered period, as in (b).

Symmetry

- In one dimension: $|F(u)| = |F(-u)|$
- In two dimensions: $|F(u, v)| = |F(-u, -v)|$



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Properties of the 2-D DFT

Symmetry

	Spatial Domain†		Frequency Domain†
1)	$f(x, y)$ real	$\Leftrightarrow$	$F^*(u, v) = F(-u, -v)$
2)	$f(x, y)$ imaginary	$\Leftrightarrow$	$F^*(-u, -v) = -F(u, v)$
3)	$f(x, y)$ real	$\Leftrightarrow$	$R(u, v)$ even; $I(u, v)$ odd
4)	$f(x, y)$ imaginary	$\Leftrightarrow$	$R(u, v)$ odd; $I(u, v)$ even
5)	$f(-x, -y)$ real	$\Leftrightarrow$	$F^*(u, v)$ complex
6)	$f(-x, -y)$ complex	$\Leftrightarrow$	$F(-u, -v)$ complex
7)	$f^*(x, y)$ complex	$\Leftrightarrow$	$F^*(-u\ -\ v)$ complex
8)	$f(x, y)$ real and even	$\Leftrightarrow$	$F(u, v)$ real and even
9)	$f(x, y)$ real and odd	$\Leftrightarrow$	$F(u, v)$ imaginary and odd
10)	$f(x, y)$ imaginary and even	$\Leftrightarrow$	$F(u, v)$ imaginary and even
11)	$f(x, y)$ imaginary and odd	$\Leftrightarrow$	$F(u, v)$ real and odd
12)	$f(x, y)$ complex and even	$\Leftrightarrow$	$F(u, v)$ complex and even
13)	$f(x, y)$ complex and odd	$\Leftrightarrow$	$F(u, v)$ complex and odd

†Recall that x, y, u, and v are *discrete* (integer) variables, with x and u in the range $[0, M - 1]$, and y, and v in the range $[0, N - 1]$. To say that a complex function is *even* means that its real *and* imaginary parts are even, and similarly for an odd complex function.

TABLE 4.1 Some symmetry properties of the 2-D DFT and its inverse. $R(u, v)$ and $I(u, v)$ are the real and imaginary parts of $F(u, v)$, respectively. The term *complex* indicates that a function has nonzero real and imaginary parts.

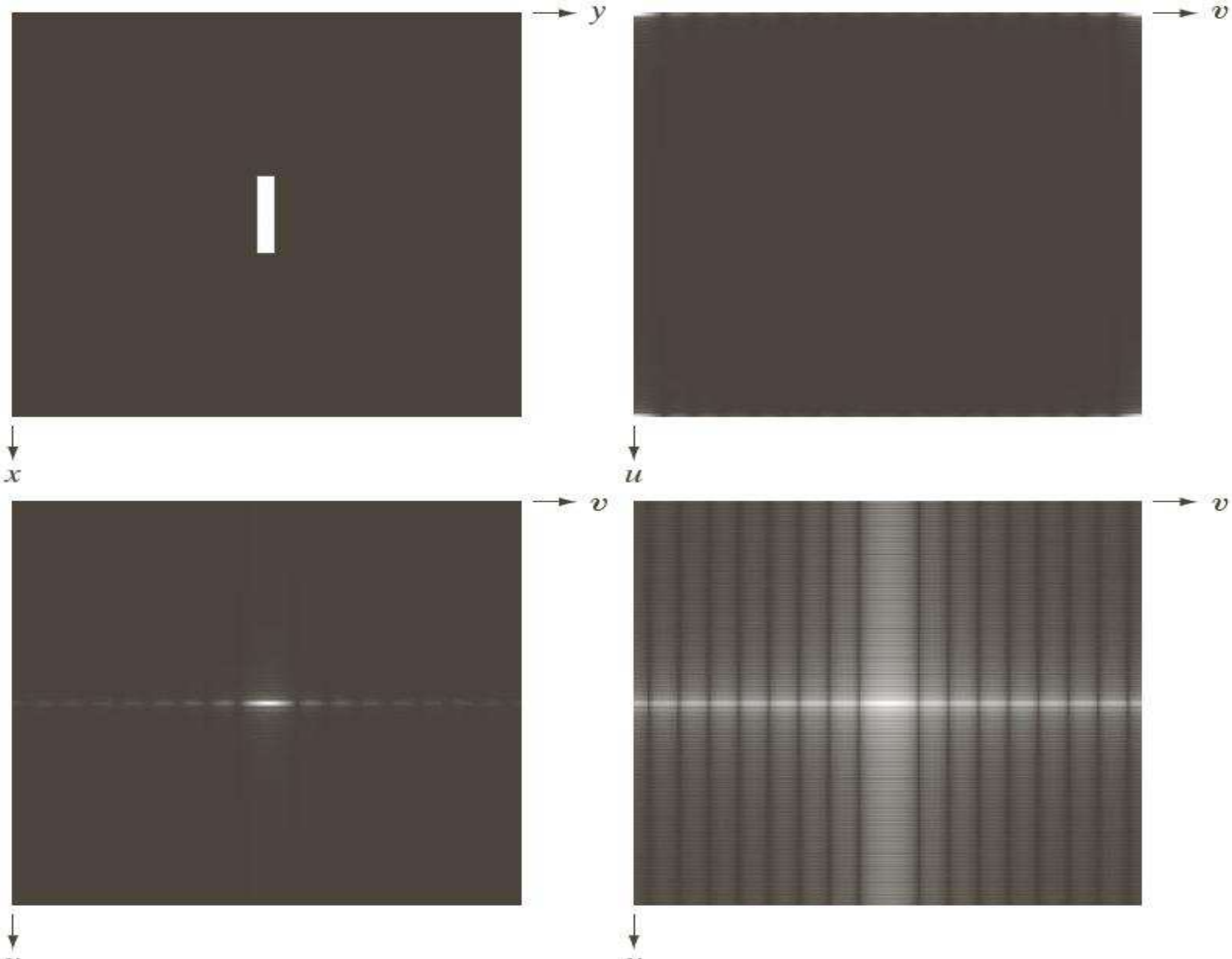
Properties of the 2-D DFT

Fourier Spectrum and Phase Angle

The FT of a real function is conjugate Symmetric

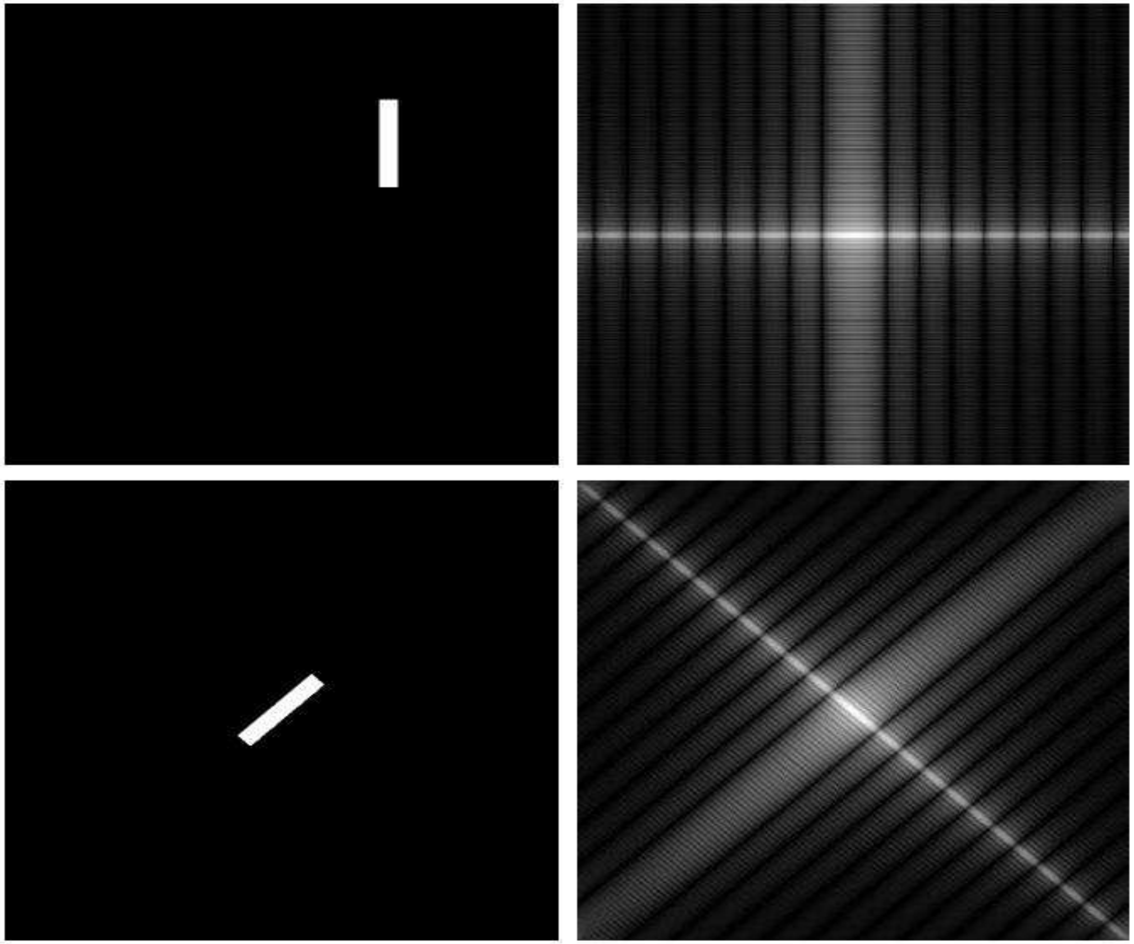
The Phase angle exhibits the following odd symmetry about the origin.

- Which indicates that zero frequency term is proportional to the average value of f(x,y)



a b
c d

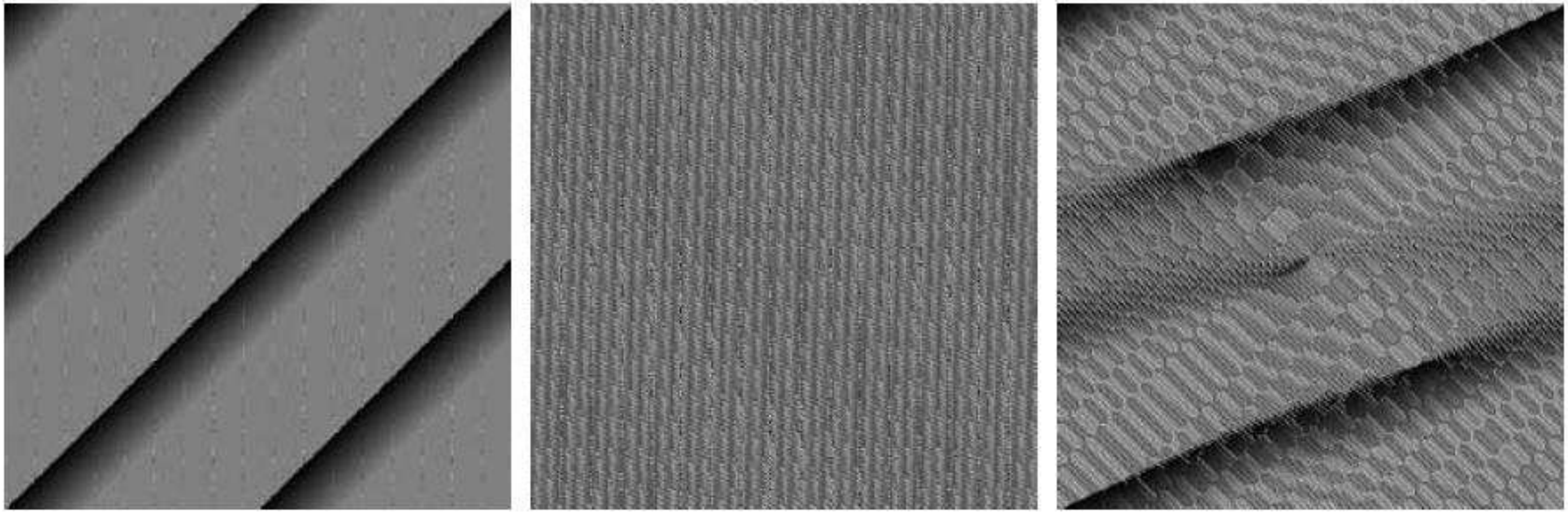
FIGURE 4.24
(a) Image.
(b) Spectrum showing bright spots in the four corners.
(c) Centered spectrum. (d) Result showing increased detail after a log transformation. The zero crossings of the spectrum are closer in the vertical direction because the rectangle in (a) is longer in that direction. The coordinate convention used throughout the book places the origin of the spatial and frequency domains at the top left.



a	b
c	d

FIGURE 4.25 (a) The rectangle in Fig. 4.24(a) translated, and (b) the corresponding spectrum. (c) Rotated rectangle, and (d) the corresponding spectrum. The spectrum corresponding to the translated rectangle is identical to the spectrum corresponding to the original image in Fig. 4.24(a).

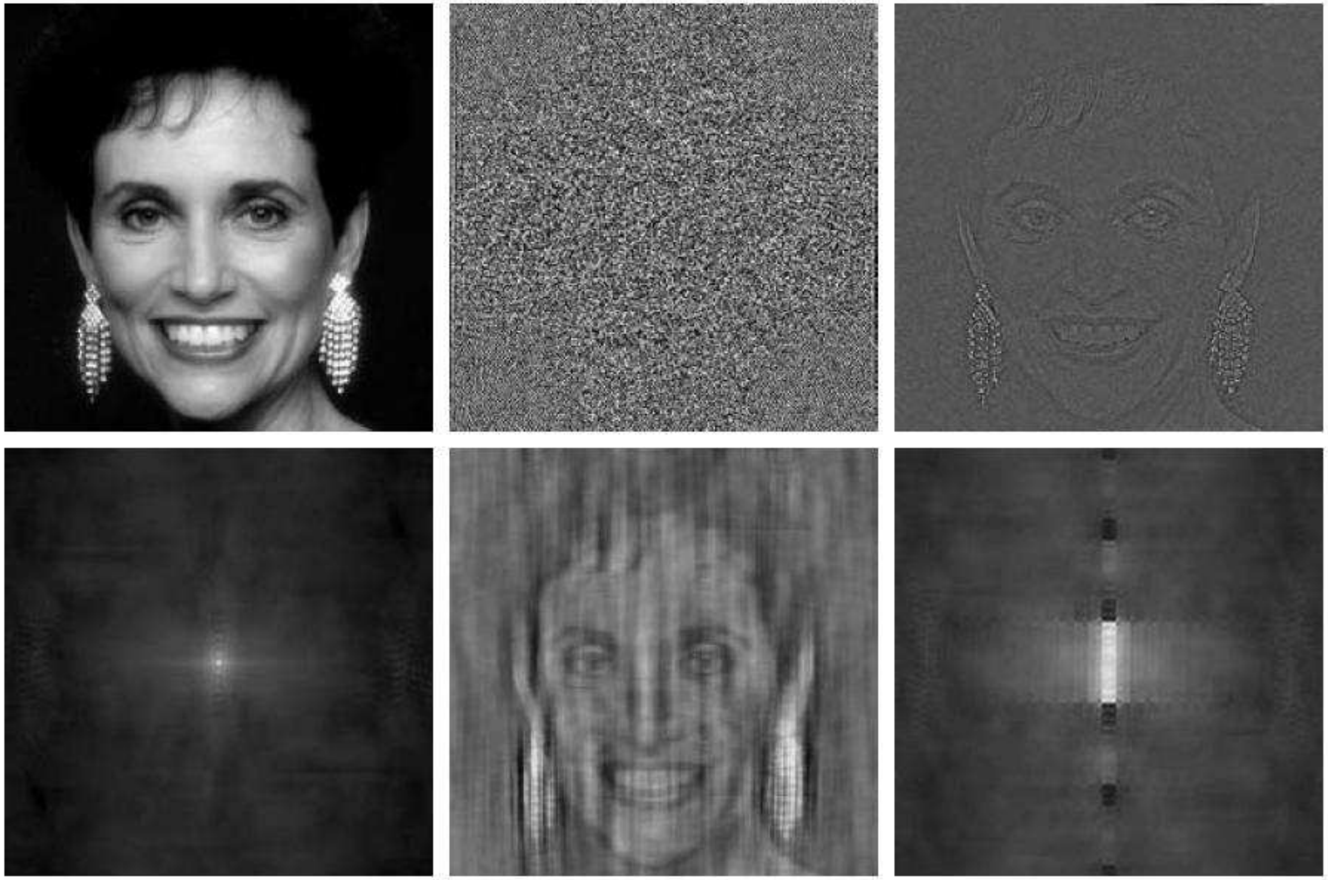
Example: Phase Angles



a b c

FIGURE 4.26 Phase angle array corresponding (a) to the image of the centered rectangle in Fig. 4.24(a), (b) to the translated image in Fig. 4.25(a), and (c) to the rotated image in Fig. 4.25(c).

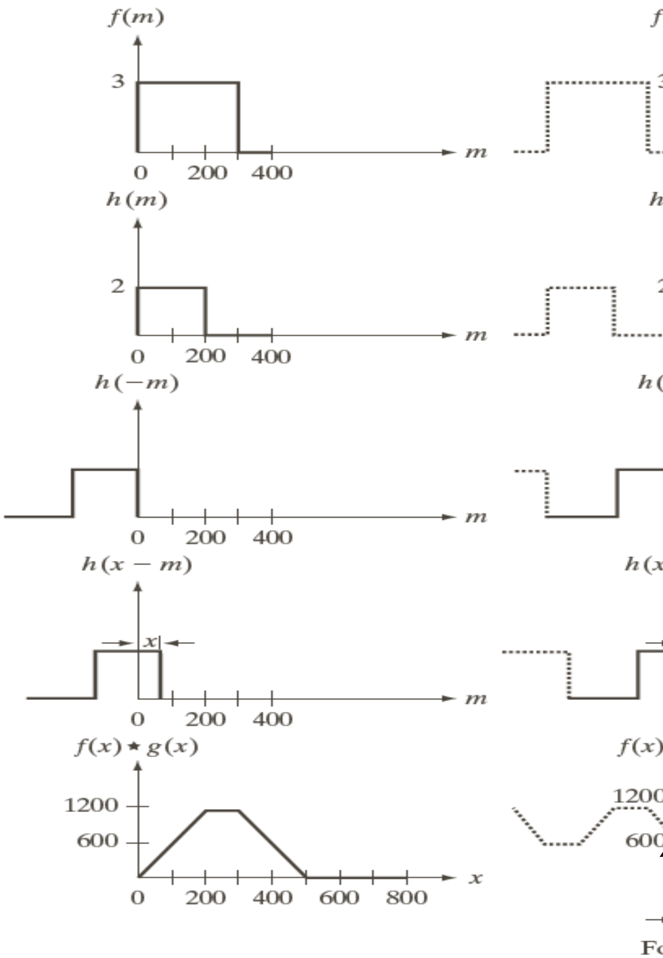
Example: Phase Angles and The Reconstructed



a	b	c
d	e	f

FIGURE 4.27 (a) Woman. (b) Phase angle. (c) Woman reconstructed using only the phase angle. (d) Woman reconstructed using only the spectrum. (e) Reconstruction using the phase angle corresponding to the woman and the spectrum corresponding to the rectangle in Fig. 4.24(a). (f) Reconstruction using the phase of the rectangle and the spectrum of the woman.

An Example of Convolution



Mirroring h about the origin

Translating the mirrored function by x

Computing the sum for each x

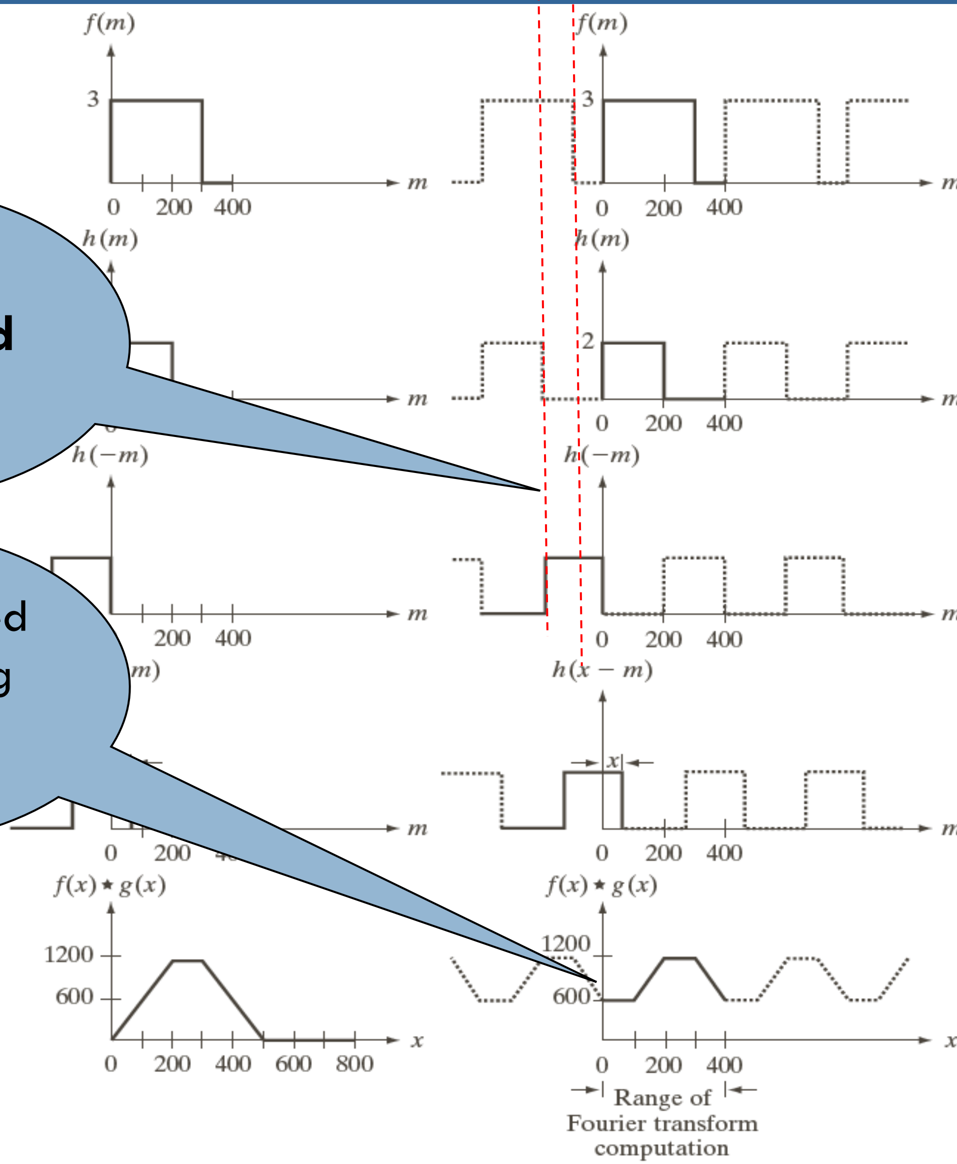
a f
b g
c h
d i
e j

FIGURE 4.28 Left column: convolution of two discrete functions obtained using the approach discussed in Section 3.4.2. The result in (e) is correct. Right column: Convolution of the same functions, but taking into account the periodicity implied by the DFT. Note in (j) how data from adjacent periods produce wraparound error, yielding an incorrect convolution result. To obtain the correct result, function padding must be used.

An Example of Convolution

It causes the **wraparound error**

It can be solved by appending zeros



a f
b g
c h
d i
e j

FIGURE 4.28 Left column: convolution of two discrete functions obtained using the approach discussed in Section 3.4.2. The result in (e) is correct. Right column: Convolution of the same functions, but taking into account the periodicity implied by the DFT. Note in (j) how data from adjacent periods produce wraparound error, yielding an incorrect convolution result. To obtain the correct result, function padding must be used.

Name	Expression(s)
1) Discrete Fourier transform (DFT) of $f(x, y)$	$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$
2) Inverse discrete Fourier transform (IDFT) of $F(u, v)$	$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$
3) Polar representation	$F(u, v) = \lvert F(u, v) \rvert e^{j\phi(u,v)}$
4) Spectrum	$\lvert F(u, v) \rvert = \left[R^2(u, v) + I^2(u, v) \right]^{1/2}$ $R = \text{Real}(F); \quad I = \text{Imag}(F)$
5) Phase angle	$\phi(u, v) = \tan^{-1}\left[\frac{I(u, v)}{R(u, v)} \right]$
6) Power spectrum	$P(u, v) = \lvert F(u, v) \rvert^2$
7) Average value	$\bar{f}(x, y) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) = \frac{1}{MN} F(0, 0)$

(Continued)

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Name	Expression(s)
8) Periodicity (k_1 and k_2 are integers)	$F(u,v) = F(u + k_1M, v) = F(u, v + k_2N) = F(u + k_1M, v + k_2N)$ $f(x,y) = f(x + k_1M, y) = f(x, y + k_2N) = f(x + k_1M, y + k_2N)$
9) Convolution	$f(x,y) \bigstar h(x,y) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)h(x-m, y-n)$
10) Correlation	$f(x,y) \star h(x,y) = \sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f^*(m,n)h(x+m, y+n)$
11) Separability	The 2-D DFT can be computed by computing 1-D DFT transforms along the rows (columns) of the image, followed by 1-D transforms along the columns (rows) of the result. See Section 4.11.1.
12) Obtaining the inverse Fourier transform using a forward transform algorithm.	$MNf^*(x,y) = \sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F^*(u,v)e^{-j2\pi(ux/M + vy/N)}$ This equation indicates that inputting $F^*(u,v)$ into an algorithm that computes the forward transform (right side of above equation) yields $MNf^*(x,y)$. Taking the complex conjugate and dividing by MN gives the desired inverse. See Section 4.11.2.

Name	DFT Pairs
1) Symmetry properties	See Table 4.1
2) Linearity	$af_1(x, y) + bf_2(x, y) \Leftrightarrow aF_1(u, v) + bF_2(u, v)$
3) Translation (general)	$f(x, y)e^{j2\pi(u_0x/M + v_0y/N)} \Leftrightarrow F(u - u_0, v - v_0)$ $f(x - x_0, y - y_0) \Leftrightarrow F(u, v)e^{-j2\pi(ux_0/M + vy_0/N)}$
4) Translation to center of the frequency rectangle, $(M/2, N/2)$	$f(x, y)(-1)^{x+y} \Leftrightarrow F(u - M/2, v - N/2)$ $f(x - M/2, y - N/2) \Leftrightarrow F(u, v)(-1)^{u+v}$
5) Rotation	$f(r, \theta + \theta_0) \Leftrightarrow F(\omega, \varphi + \theta_0)$ $x = r\cos\theta \quad y = r\sin\theta \quad u = \omega\cos\varphi \quad v = \omega\sin\varphi$
6) Convolution theorem†	$f(x, y) \star h(x, y) \Leftrightarrow F(u, v)H(u, v)$ $f(x, y)h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$

(*Continued*)

Name	DFT Pairs
7) Correlation theorem†	$f(x, y) \star h(x, y) \Leftrightarrow F^*(u, v)\, H(u, v)$ $f^*(x, y) h(x, y) \Leftrightarrow F(u, v) \star H(u, v)$
8) Discrete unit impulse	$\delta(x, y) \Leftrightarrow 1$
9) Rectangle	$\text{rect}[a, b] \Leftrightarrow ab \dfrac{\sin(\pi ua)}{(\pi ua)} \dfrac{\sin(\pi vb)}{(\pi vb)} e^{-j\pi(ua+vb)}$
10) Sine	$\sin(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $j\frac{1}{2}\left[\delta(u + Mu_0, v + Nv_0) - \delta(u - Mu_0, v - Nv_0)\right]$
11) Cosine	$\cos(2\pi u_0 x + 2\pi v_0 y) \Leftrightarrow$ $\frac{1}{2}\left[\delta(u + Mu_0, v + Nv_0) + \delta(u - Mu_0, v - Nv_0)\right]$
The following Fourier transform pairs are derivable only for continuous variables, denoted as before by t and z for spatial variables and by μ and ν for frequency variables. These results can be used for DFT work by sampling the continuous forms.	
12) *Differentiation* (The expressions on the right assume that $f(\pm\infty, \pm\infty) = 0$.)	$\left(\frac{\partial}{\partial t}\right)^m \left(\frac{\partial}{\partial z}\right)^n f(t, z) \Leftrightarrow (j2\pi\mu)^m (j2\pi\nu)^n F(\mu, \nu)$ $\dfrac{\partial^m f(t, z)}{\partial t^m} \Leftrightarrow (j2\pi\mu)^m F(\mu, \nu); \dfrac{\partial^n f(t, z)}{\partial z^n} \Leftrightarrow (j2\pi\nu)^n F(\mu, \nu)$
13) *Gaussian*	$A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)} \Leftrightarrow Ae^{-(\mu^2+\nu^2)/2\sigma^2}$ (A is a constant)

†Assumes that the functions have been extended by zero padding. Convolution and correlation are associative, commutative, and distributive.

Relation Between Spatial and Frequency Resolutions

where

Δx = spatial resolution in x direction

Δy = spatial resolution in y direction

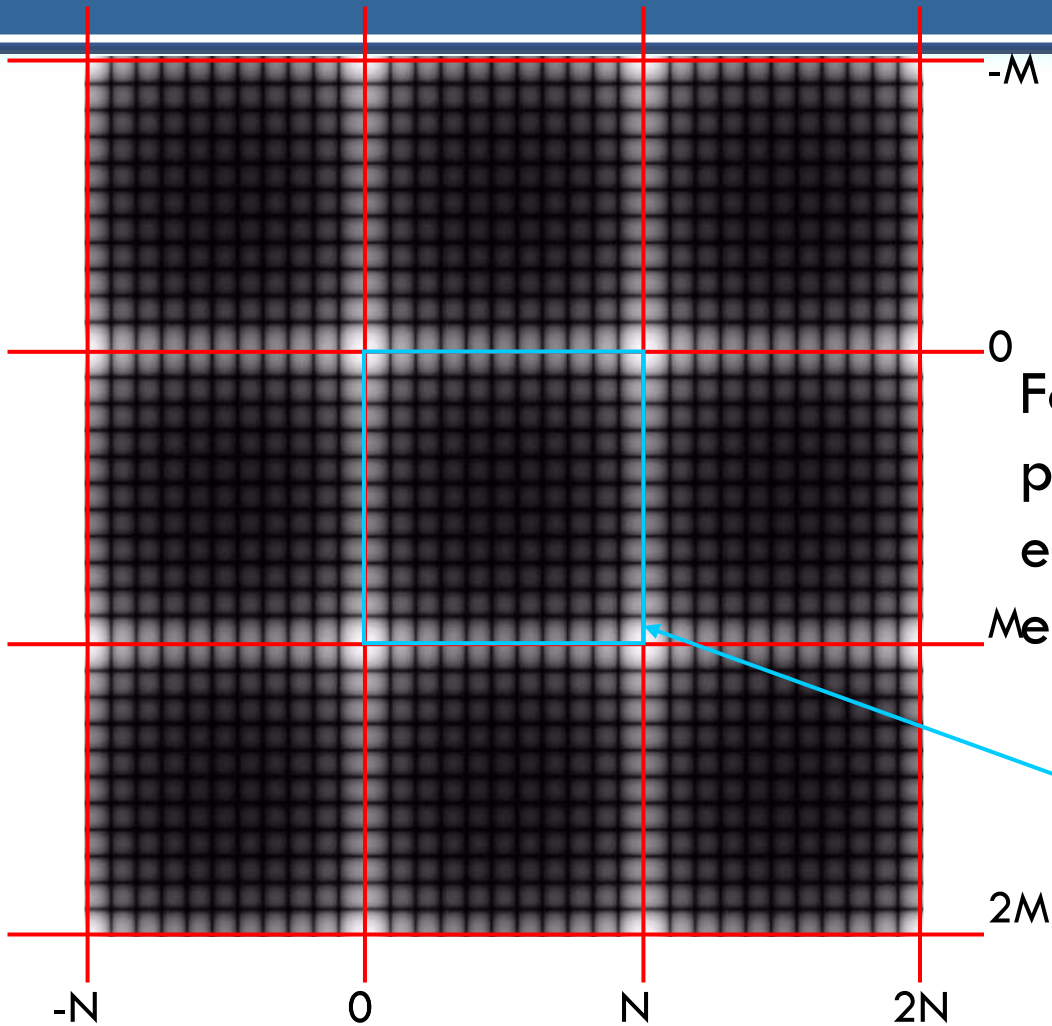
(Δx and Δy are pixel width and height.)

Δu = frequency resolution in x direction

Δv = frequency resolution in y direction

N,M = image width and height

Periodicity of 2-D DFT



2-D DFT:

For an image of size NxM pixels, its 2-D DFT repeats itself every N points in x-direction and every M points in y-direction.

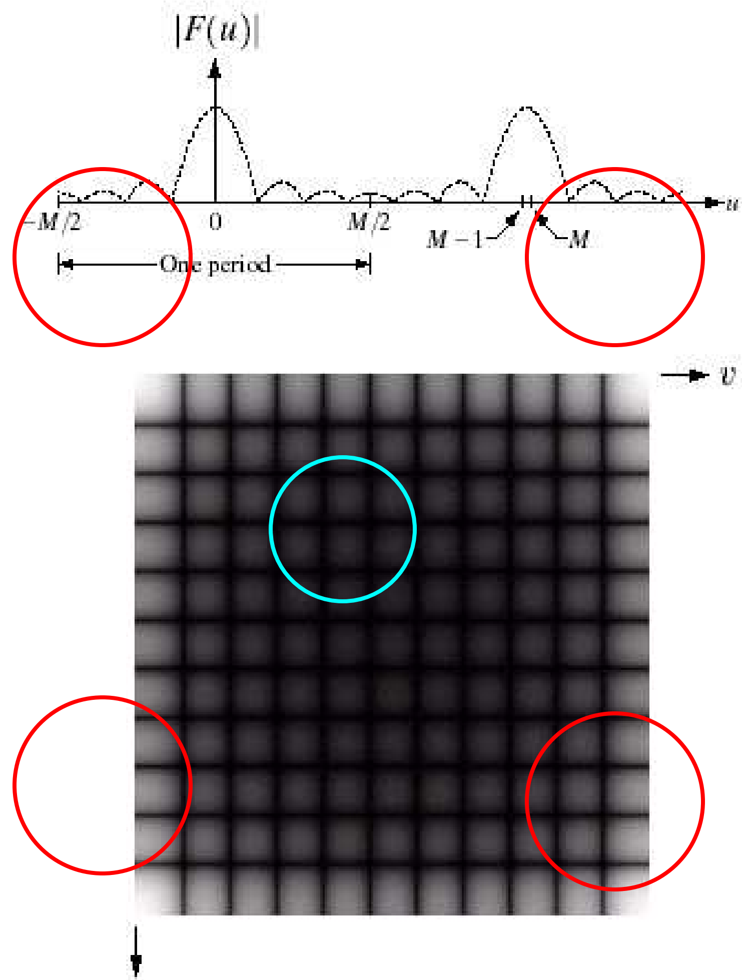
$g(x,y)$

We display only in this range

F(*u*,*v*) has low frequency areas at corners of the image while high frequency areas are at the center of the image which is inconvenient to interpret.

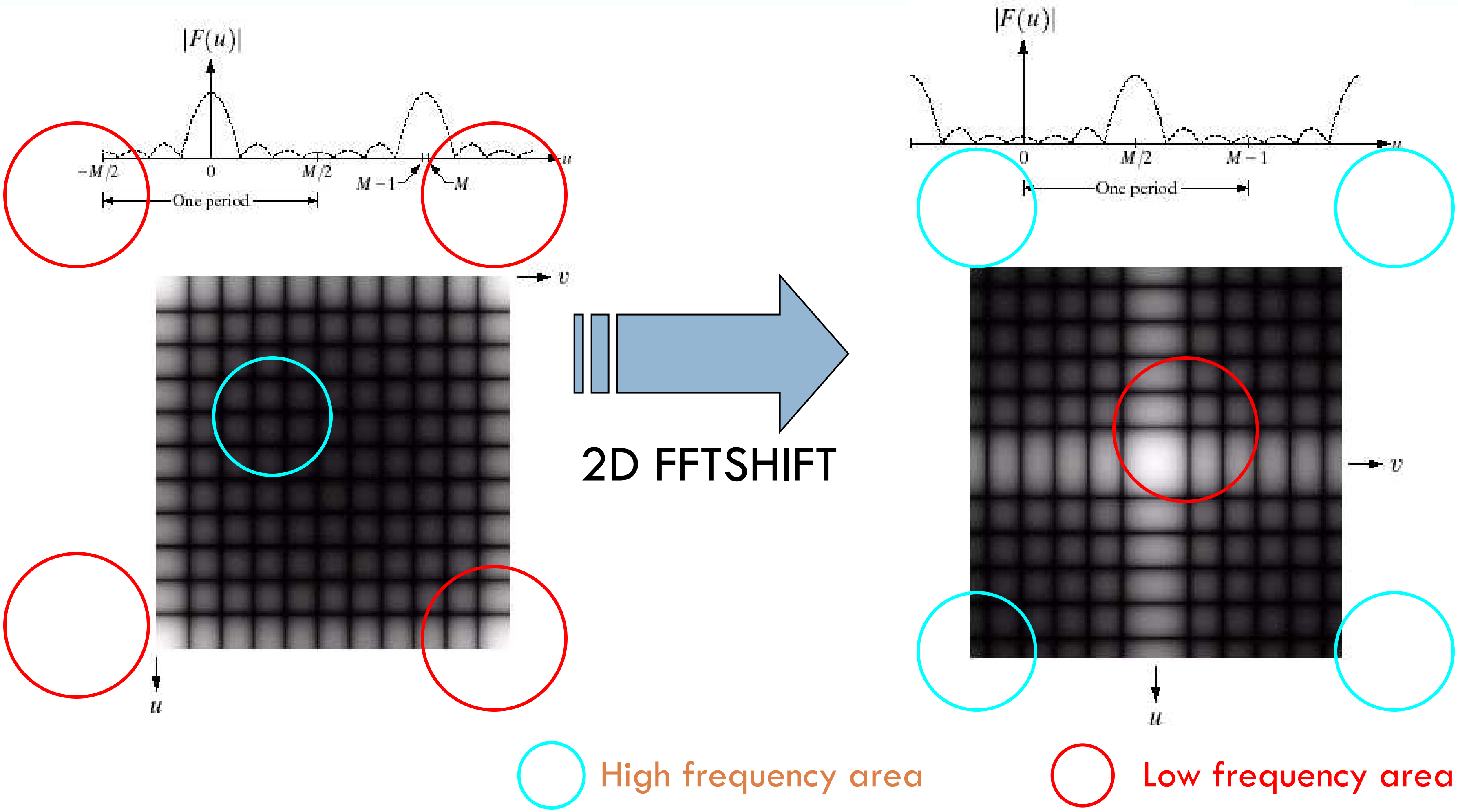
High frequency area

Low frequency area

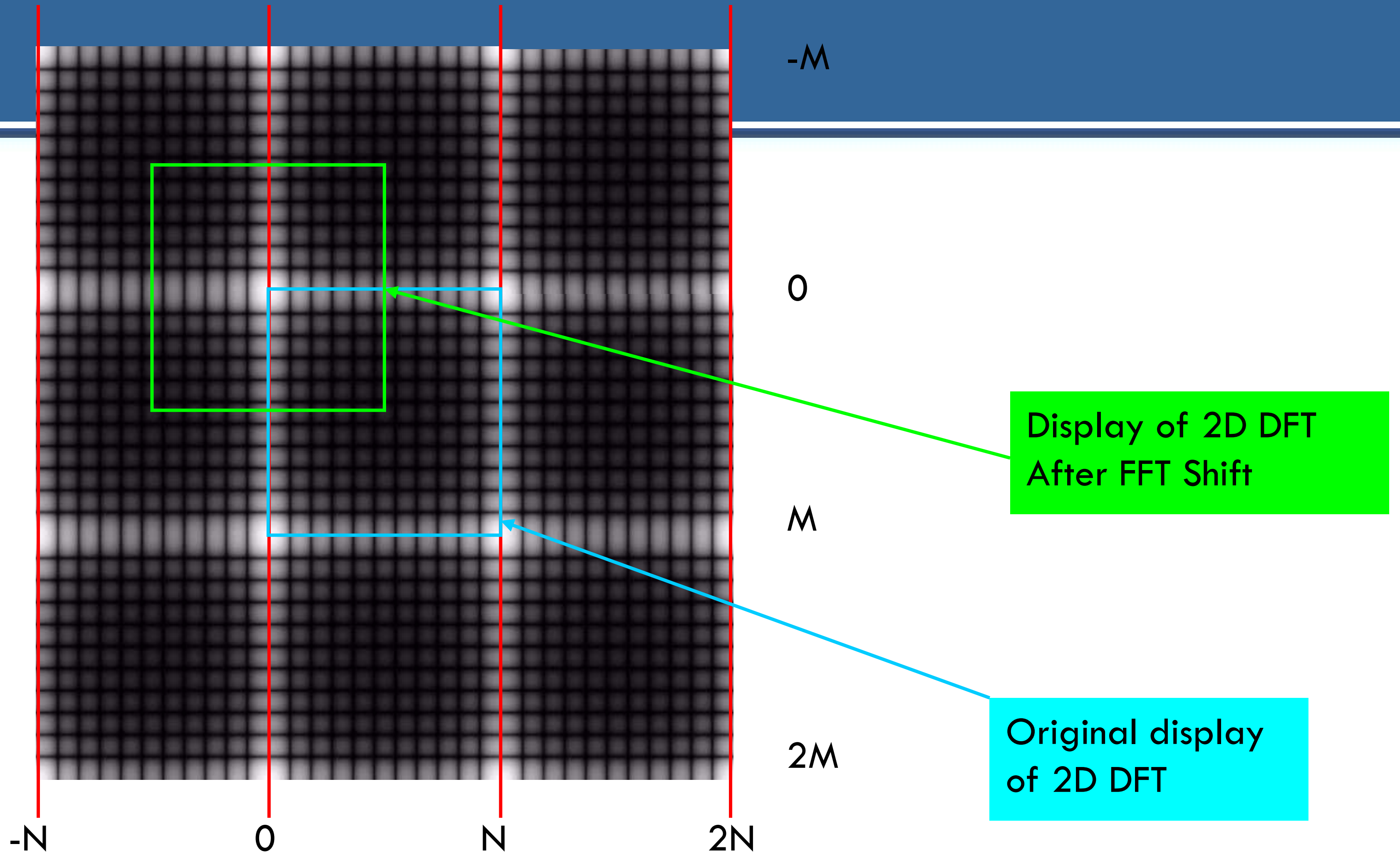


2-D FFT Shift : Better Display of 2-D DFT

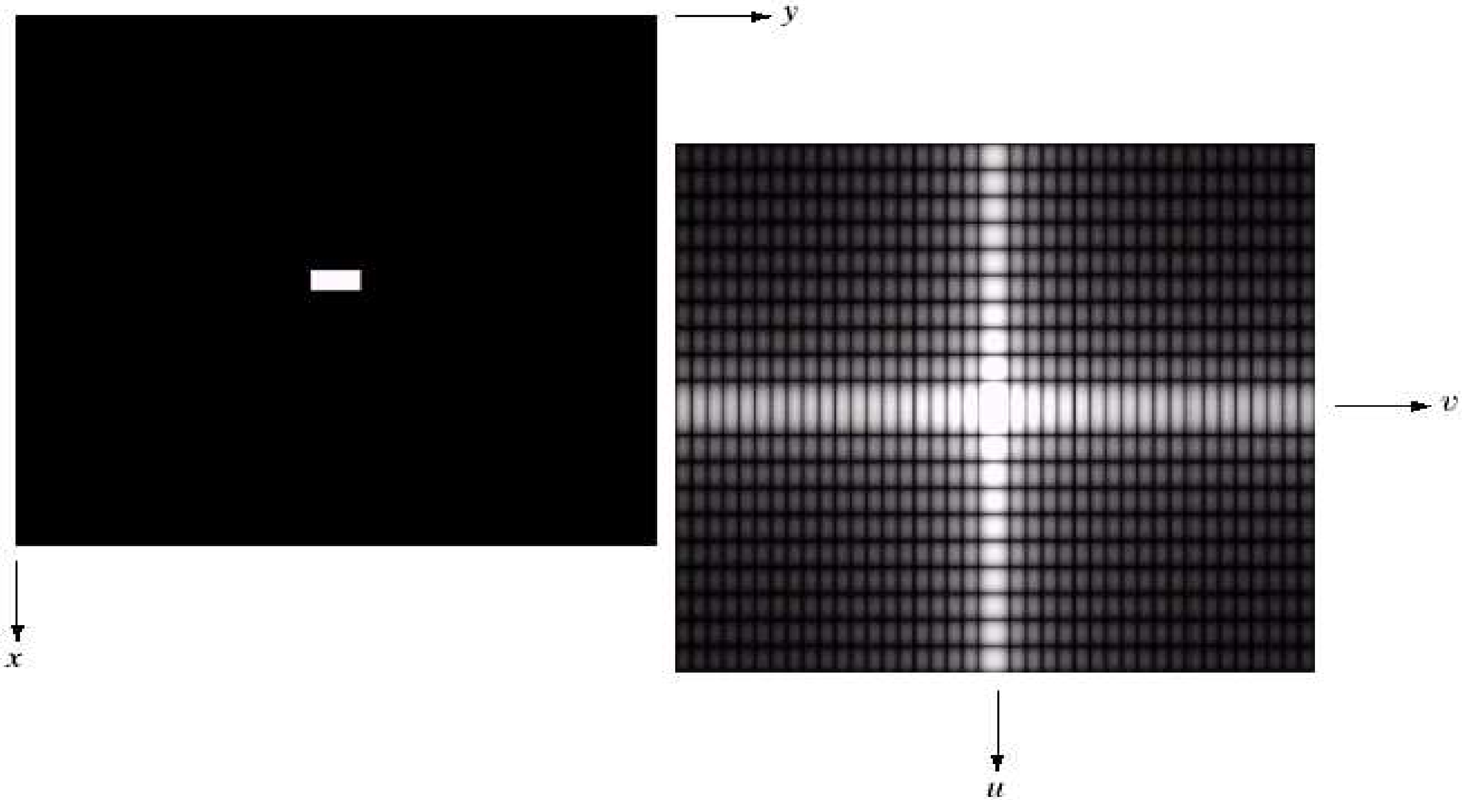
2-D FFT Shift is a MATLAB function: Shift the zero frequency of $F(u,v)$ to the center of an image.



2-D FFT Shift (cont.) : How it works



Example of 2-D DFT

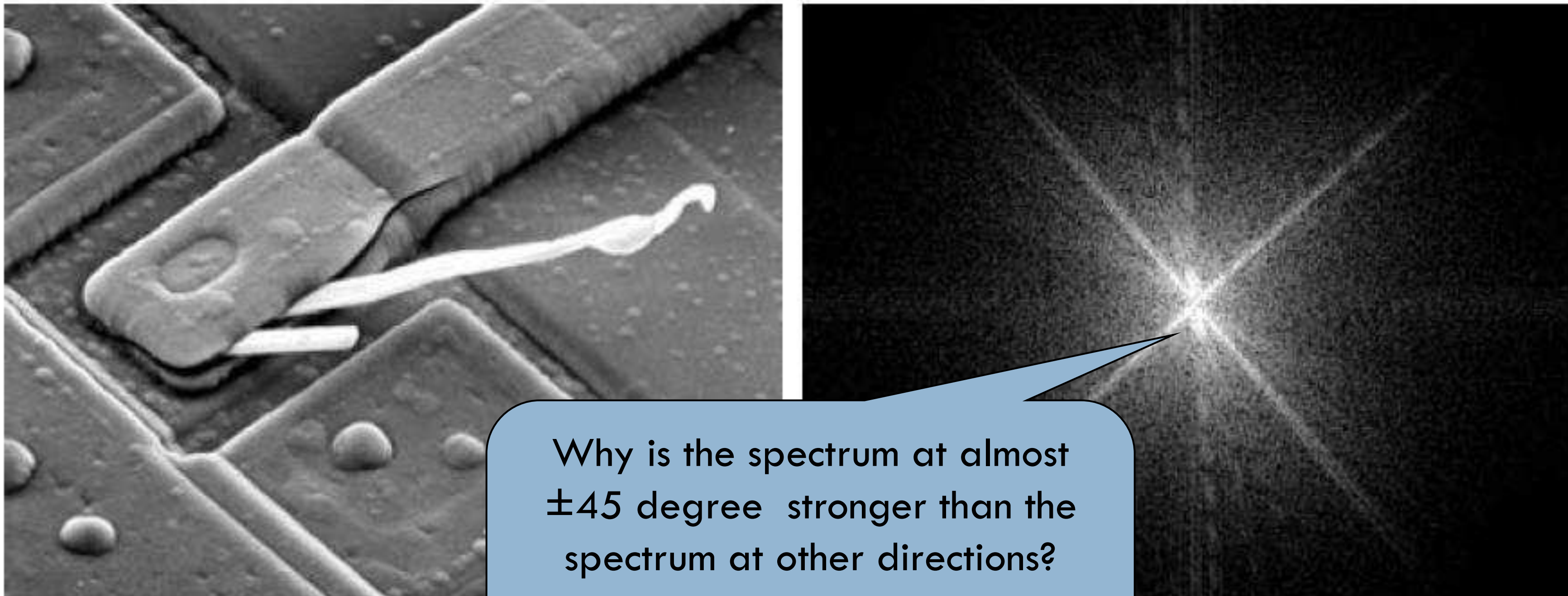


a b

FIGURE 4.3 (a) Image of a 20 × 40 white rectangle on a black background of size 512 × 512 pixels. (b) Centered Fourier spectrum shown after application of the log transformation given in Eq. (3.2-2). Compare with Fig. 4.2.

Notice that the longer the time domain signal,
The shorter its Fourier transform

The Basic Filtering in the Frequency Domain



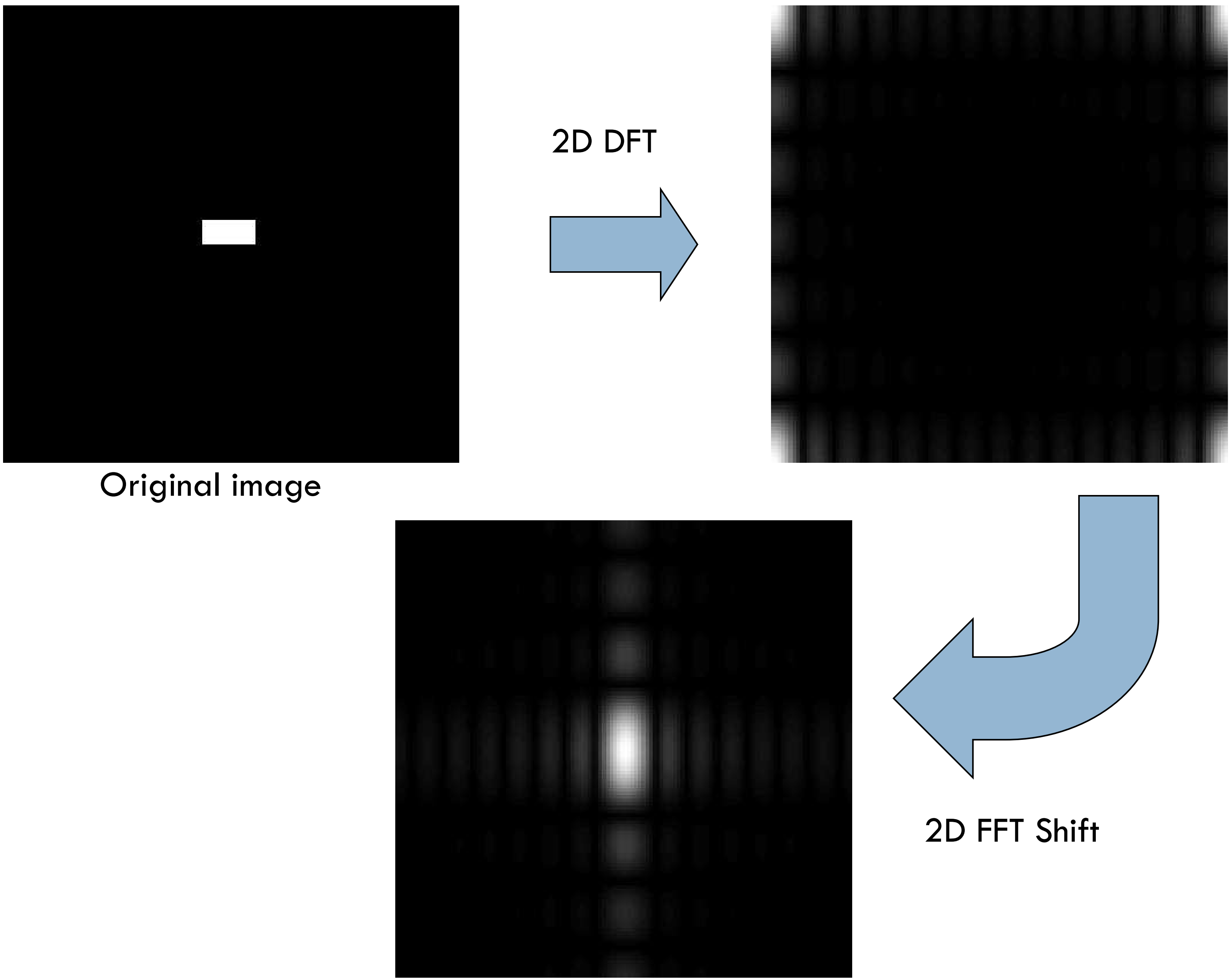
Why is the spectrum at almost ±45 degree stronger than the spectrum at other directions?

a b

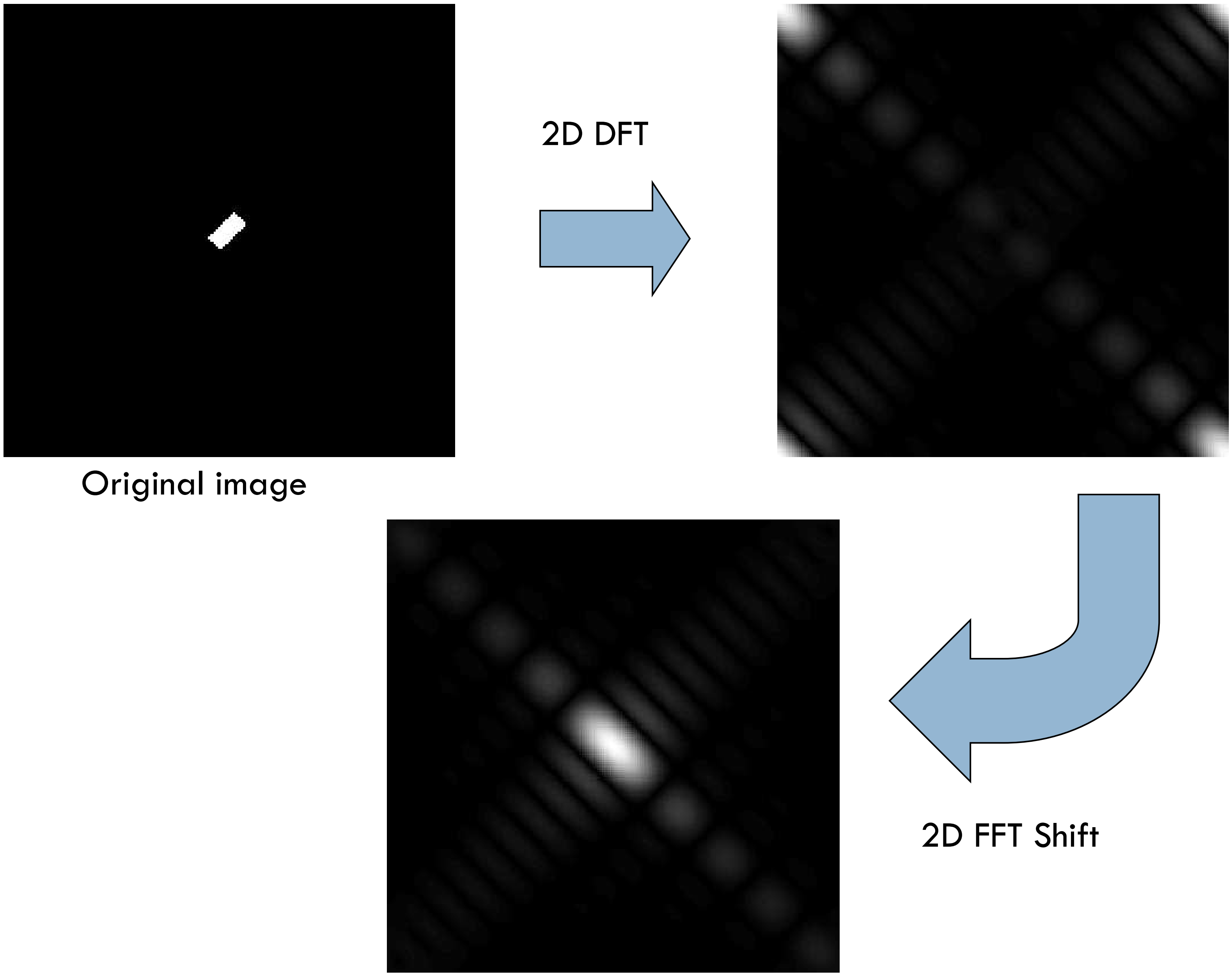
FIGURE 4.29 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

Notice that direction of an object in spatial image and Its Fourier transform are orthogonal to each other.

Example of 2-D DFT



Example of 2-D DFT



Basic steps for filtering in the Frequency domain

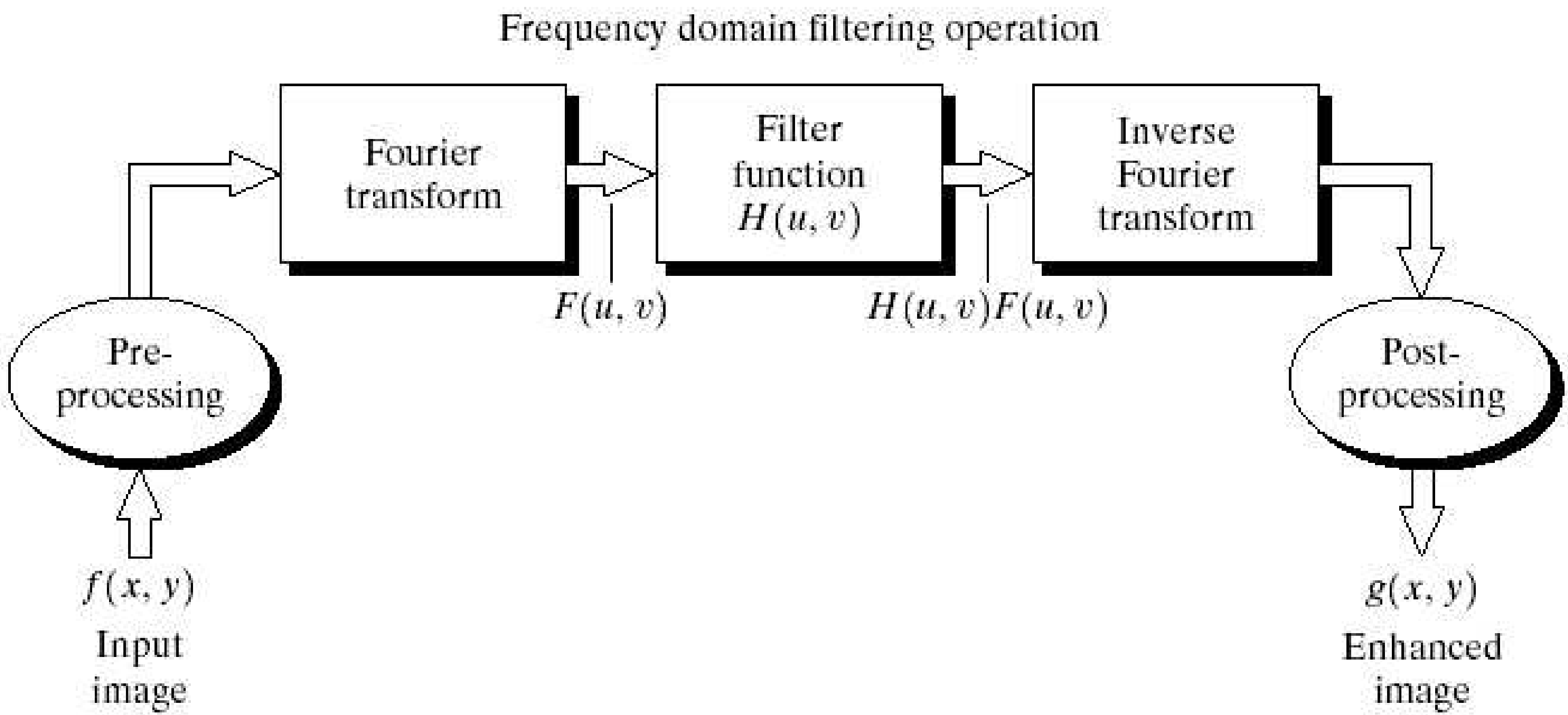


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Basics of Low Pass Filters in the Frequency Domain:

Low pass filter: A filter that attenuates high frequencies while passing the low frequencies.
Low frequencies represent the gray-level appearance of an image over smooth areas.

High pass filter: A filter that attenuates low frequencies while passing the high frequencies.
High frequencies represents the details such as edges and noise.

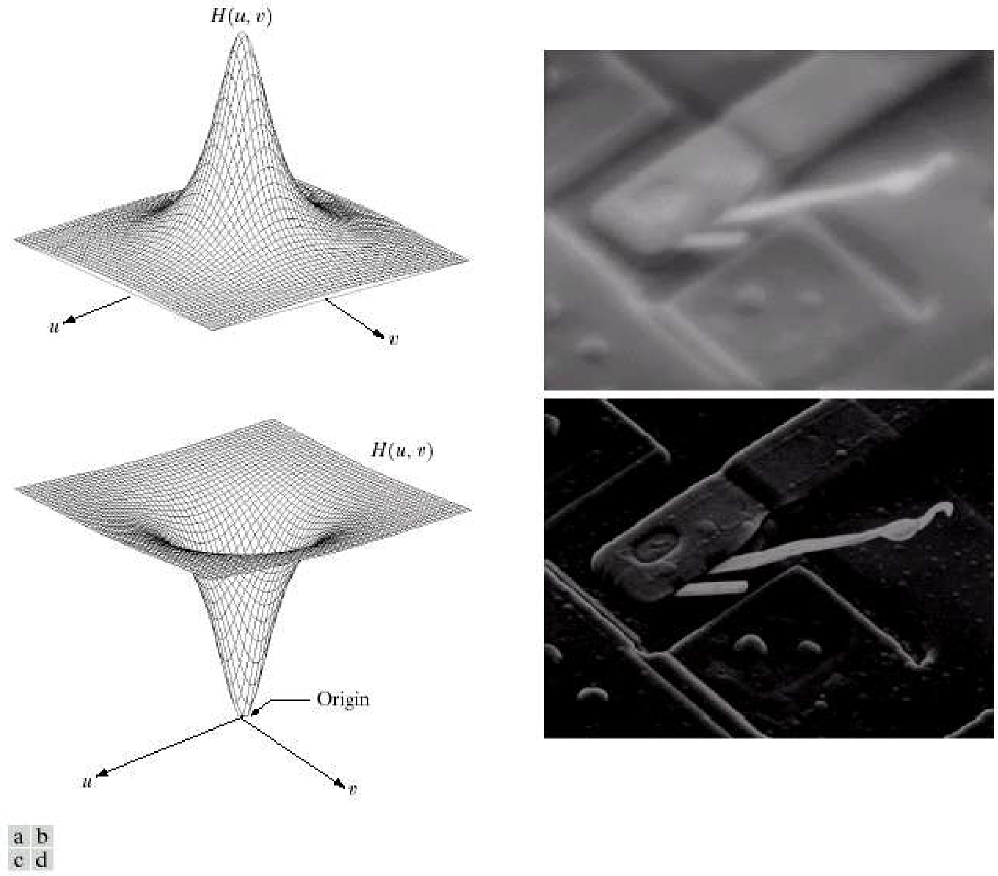
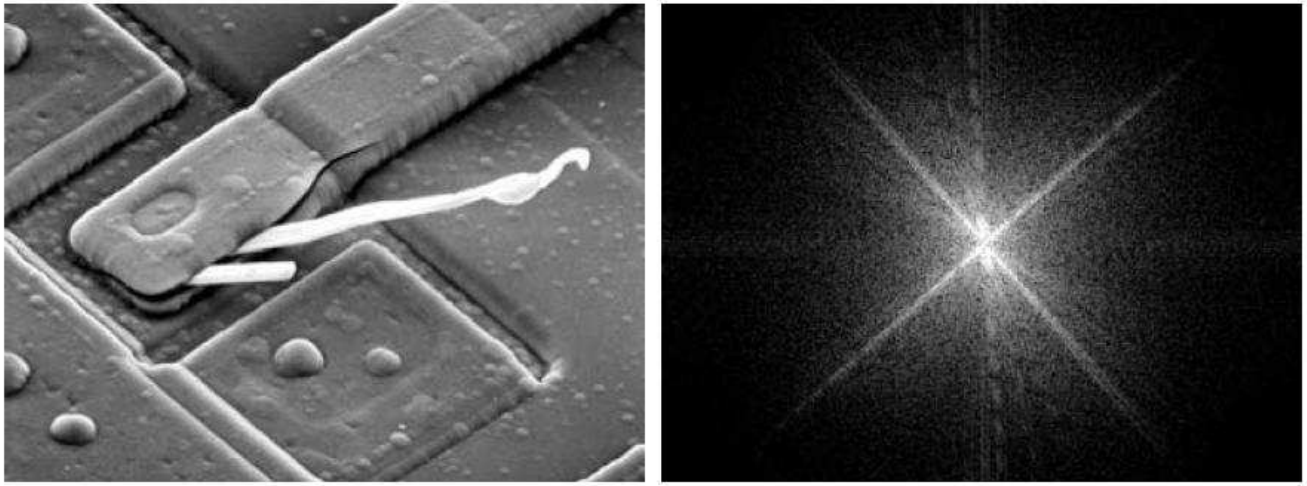


FIGURE 4.7 (a) A two-dimensional lowpass filter function. (b) Result of lowpass filtering the image in Fig. 4.4(a). (c) A two-dimensional highpass filter function. (d) Result of highpass filtering the image in Fig. 4.4(a).

The Basic Filtering in the Frequency Domain

- In a filter $H(u,v)$ that is 0 at the center of the transform and 1 elsewhere, what's the output image?



a b

FIGURE 4.29 (a) SEM image of a damaged integrated circuit. (b) Fourier spectrum of (a). (Original image courtesy of Dr. J. M. Hudak, Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, Canada.)

The Basic Filtering in the Frequency Domain

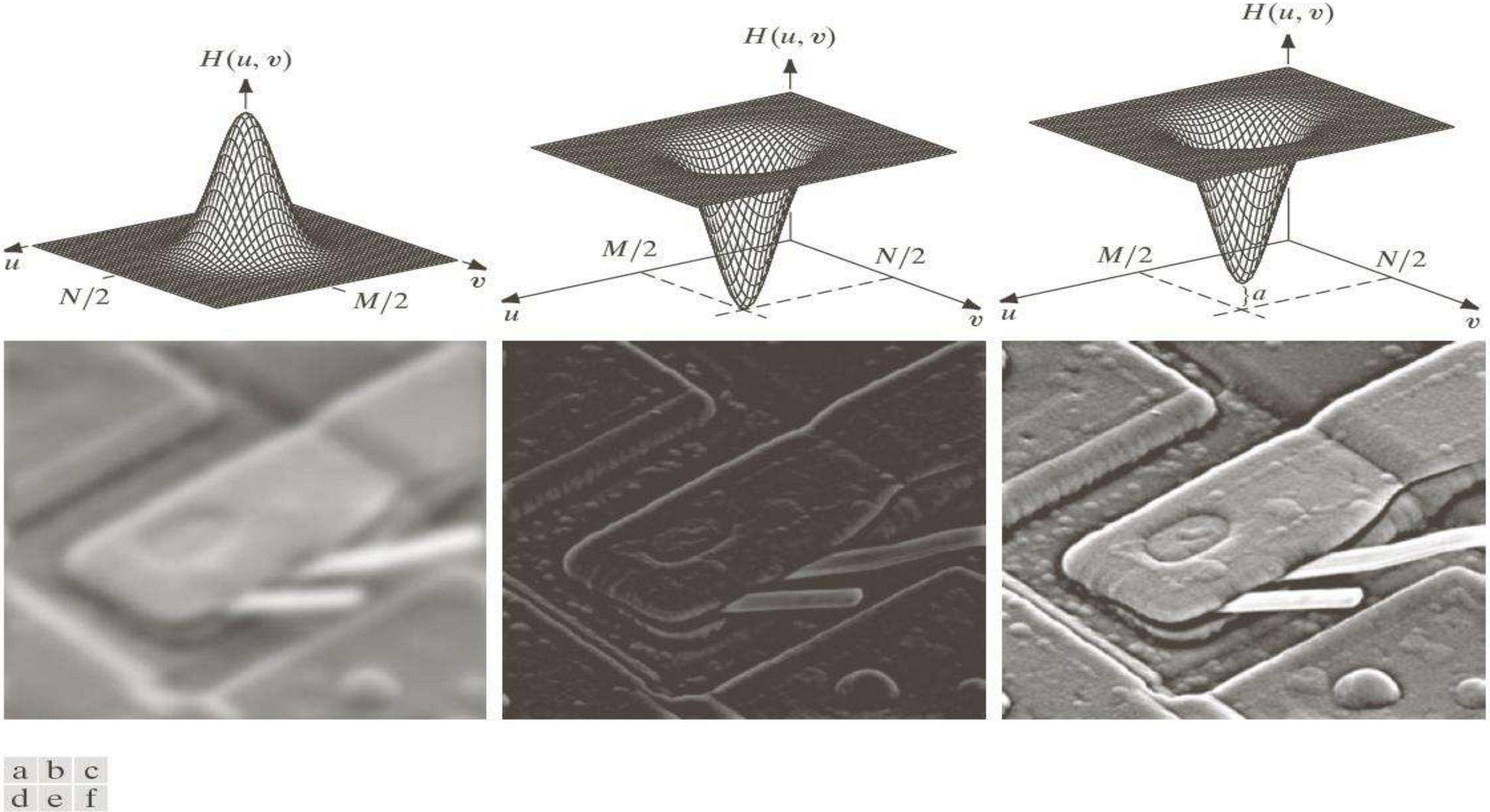


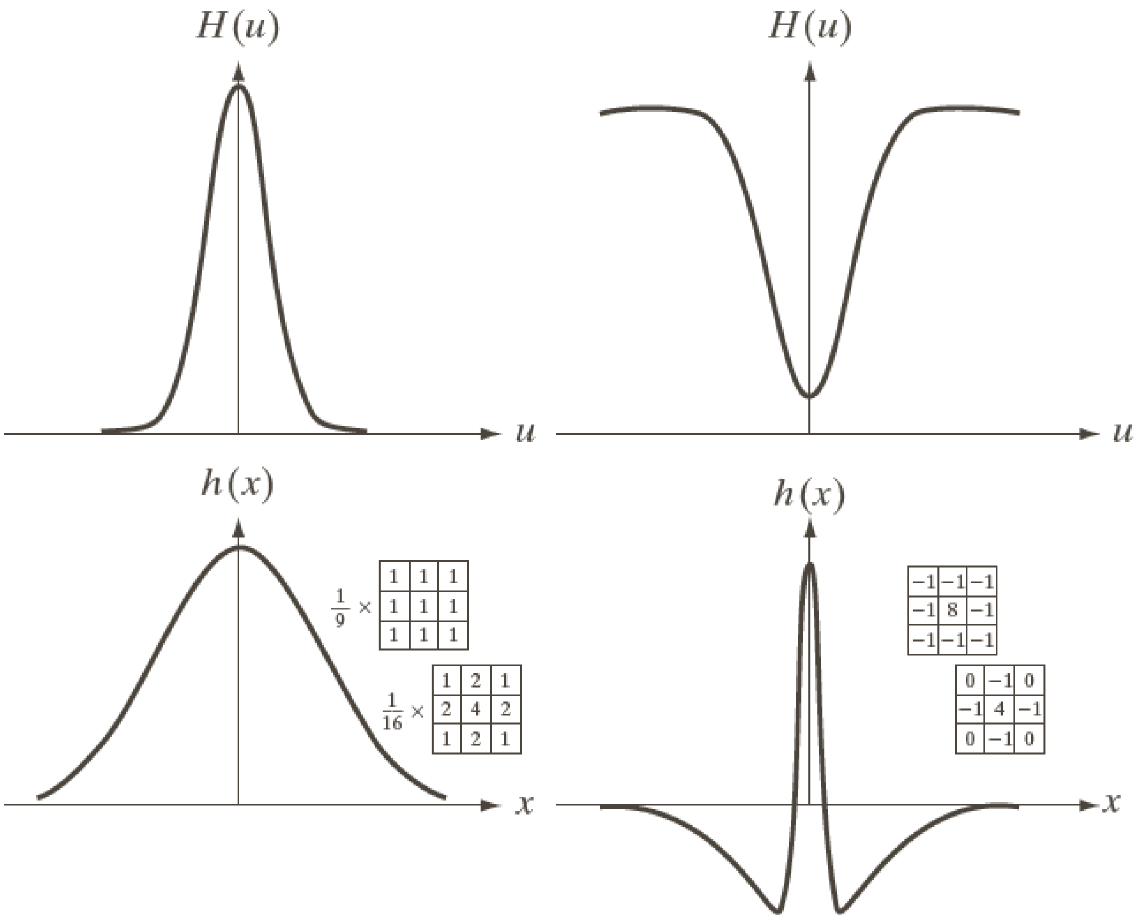
FIGURE 4.31 Top row: frequency domain filters. Bottom row: corresponding filtered images obtained using Eq. (4.7-1). We used $a = 0.85$ in (c) to obtain (f) (the height of the filter itself is 1). Compare (f) with Fig. 4.29(a).

The Basic Filtering in the Frequency Domain

a b c

FIGURE 4.32 (a) A simple image. (b) Result of blurring with a Gaussian lowpass filter without padding. (c) Result of lowpass filtering with padding. Compare the light area of the vertical edges in (b) and (c).

Correspondence Between Filtering in the Spatial and Frequency Domains (3)

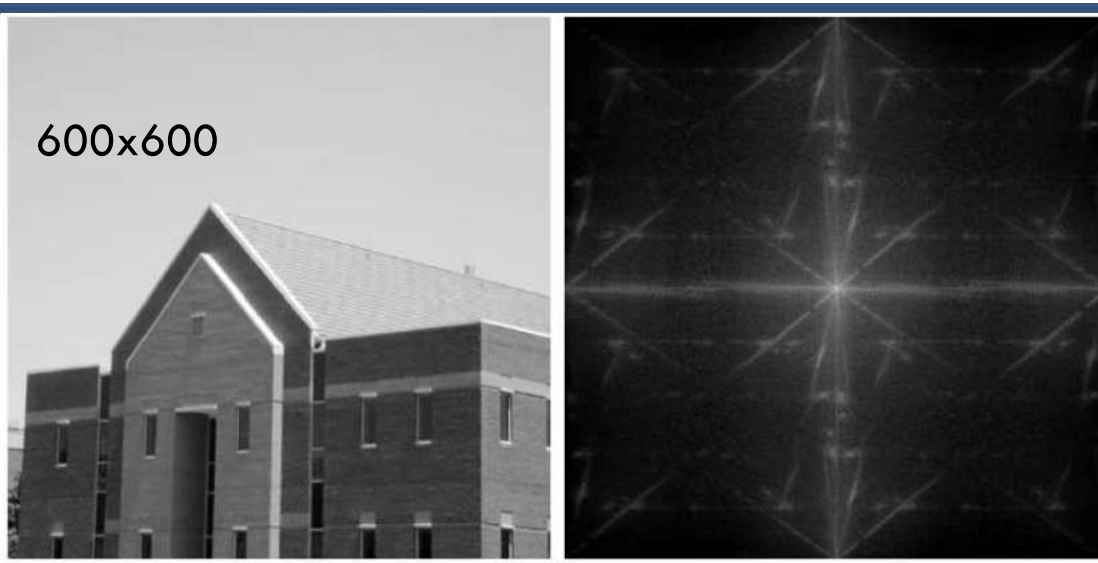


a c
b d

FIGURE 4.37 (a) A 1-D Gaussian lowpass filter in the frequency domain. (b) Spatial lowpass filter corresponding to (a). (c) Gaussian highpass filter in the frequency domain. (d) Spatial highpass filter corresponding to (c). The small 2-D masks shown are spatial filters we used in Chapter 3.

- One very useful property of the Gaussian function is that both it and its Fourier transform are real valued; there are no complex values associated with them.
- In addition, the values are always positive. So, if we convolve an image with a Gaussian function, there will never be any negative output values to deal with.
- There is also an important relationship between the widths of a Gaussian function and its Fourier transform. If we make the width of the function smaller, the width of the Fourier transform gets larger. This is controlled by the variance parameter σ^2 in the equations.
- These properties make the Gaussian filter very useful for lowpass filtering an image. The amount of blur is controlled by σ^2. It can be implemented in either the spatial or frequency domain.
- Other filters besides lowpass can also be implemented by using two different sized Gaussian functions.

Correspondence Between Filtering in the Spatial and Frequency Domains: Example

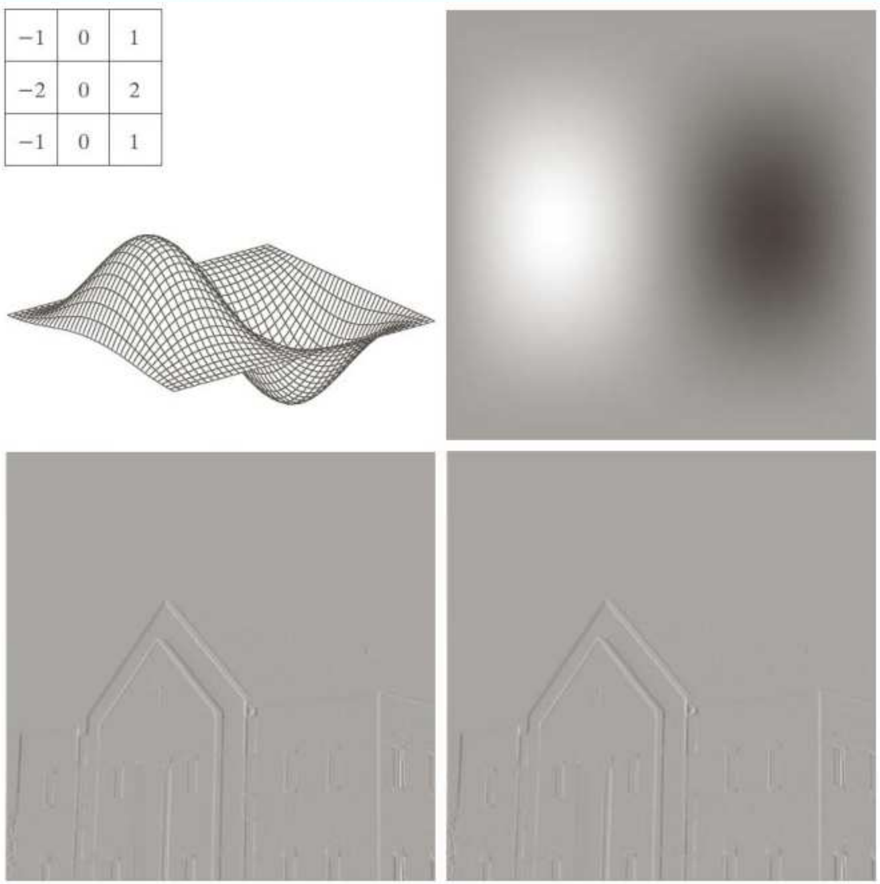


a b

FIGURE 4.38
(a) Image of a building, and (b) its spectrum.

In this example, we start with a spatial mask and show how to generate its corresponding filter in the frequency domain. Then, we compare the filtering results obtained using frequency domain and spatial techniques. We use the 3x3 Sobel vertical edge detector. The left one is a 600x600 pixel image, and its spectrum is shown on the right.

Correspondence Between Filtering in the Spatial and Frequency Domains: Example



−1	0	1
−2	0	2
−1	0	1

a b
c d

FIGURE 4.39 (a) A spatial mask and perspective plot of its corresponding frequency domain filter. (b) Filter shown as an image. (c) Result of filtering Fig. 4.38(a) in the frequency domain with the filter in (b). (d) Result of filtering the same image with the spatial filter in (a). The results are identical.

In this example, we start with a spatial mask and show how to generate its corresponding filter in the frequency domain. Then, we compare the filtering results obtained using frequency domain and spatial techniques. We use the 3x3 Sobel vertical edge detector. The left one is a 600x600 pixel image, and its spectrum is shown on the right.

Generate H(u,v)

3.Smoothing Frequency Domain Filters

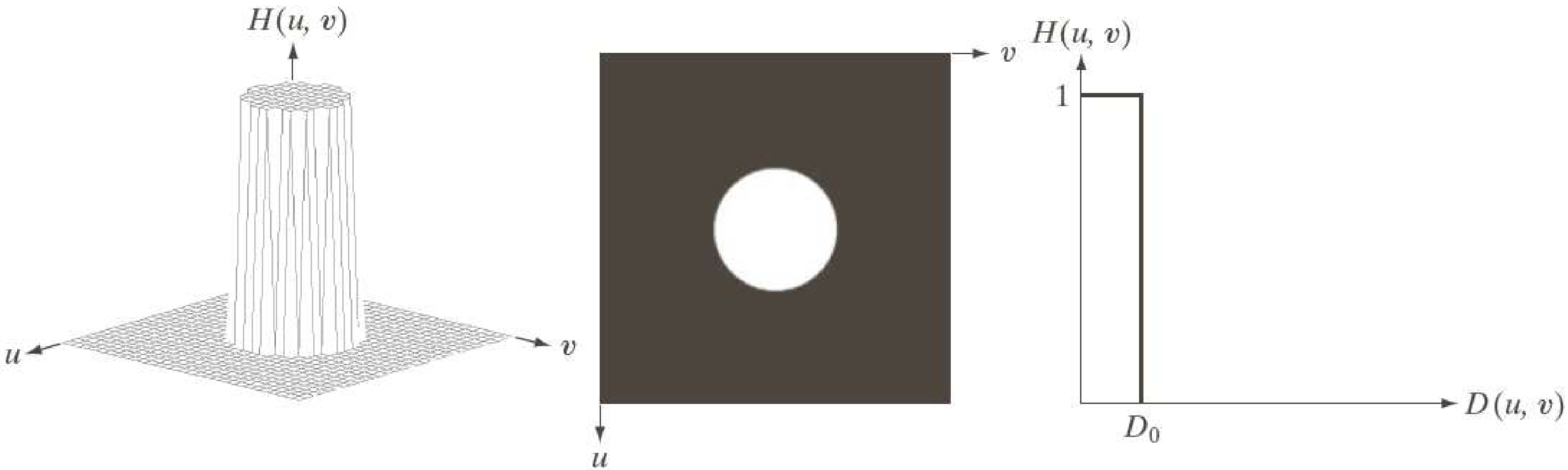
- The basic model for filtering in the frequency domain

where $F(u,v)$: the Fourier transform of the image to be smoothed
$H(u,v)$: a filter transfer function

- Smoothing is fundamentally a lowpass operation in the frequency domain.
- There are several standard forms of lowpass filters (LPF).
  - Ideal lowpass filter
  - Butterworth lowpass filter
  - Gaussian lowpass filter

Image Smoothing Using Filter Domain Filters: ILPF

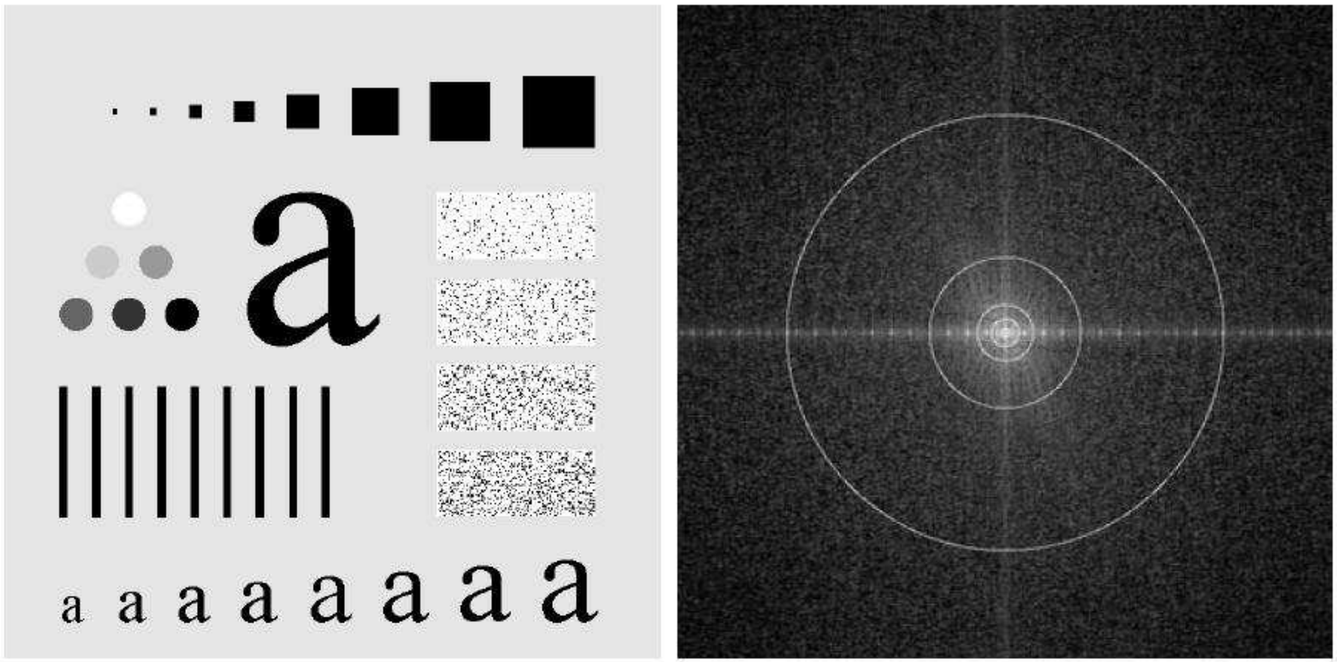
Image Smoothing Using Filter Domain Filters: ILPF



a b c

FIGURE 4.40 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross section.

ILPF Filtering Example



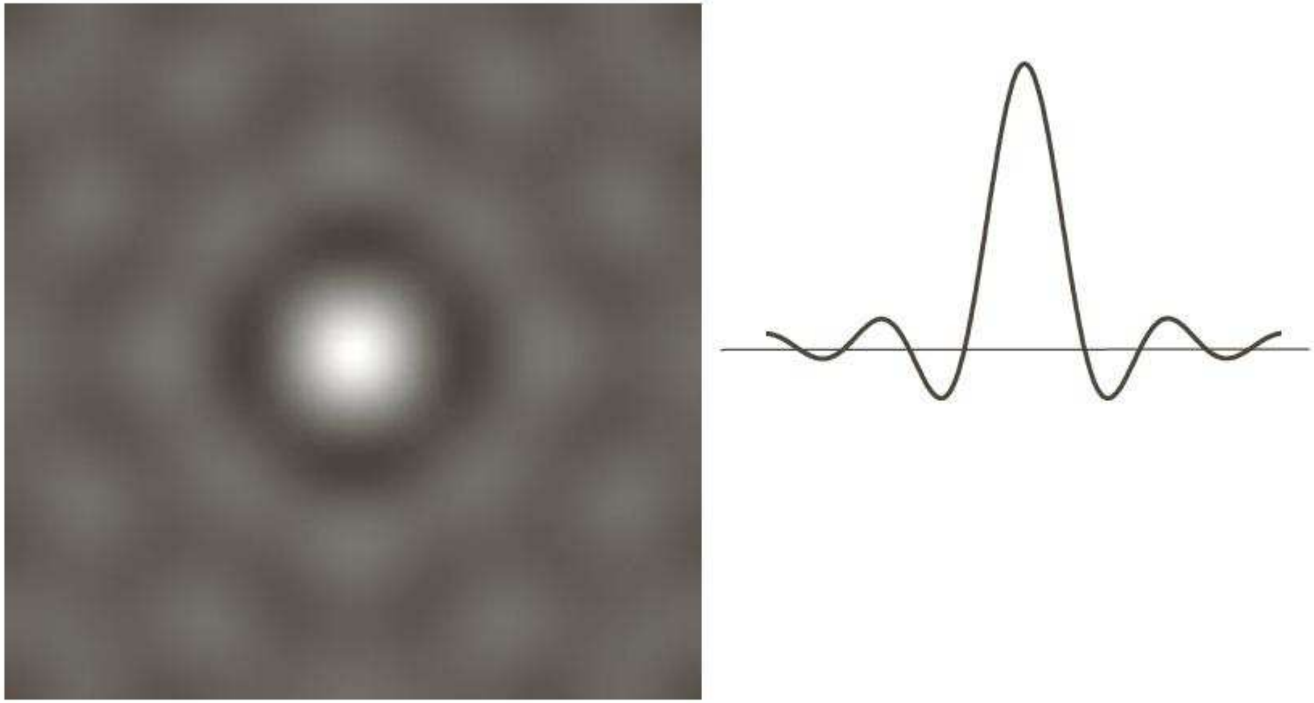
a b

FIGURE 4.41 (a) Test pattern of size 688 × 688 pixels, and (b) its Fourier spectrum. The spectrum is double the image size due to padding but is shown in half size so that it fits in the page. The superimposed circles have radii equal to 10, 30, 60, 160, and 460 with respect to the full-size spectrum image. These radii enclose 87.0, 93.1, 95.7, 97.8, and 99.2% of the padded image power, respectively.

ILPF Filtering Example

FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

The Spatial Representation of ILPF



a b

FIGURE 4.43 (a) Representation in the spatial domain of an ILPF of radius 5 and size 1000×1000. (b) Intensity profile of a horizontal line passing through the center of the image.

Ideal Low pass Filters
Another Example

Figure 4.13 (a) A frequency-domain ILPF of radius 5. (b) Corresponding spatial filter. (c) Five impulses in the spatial domain, simulating the values of five pixels. (d) Convolution of (b) and (c) in the spatial domain.

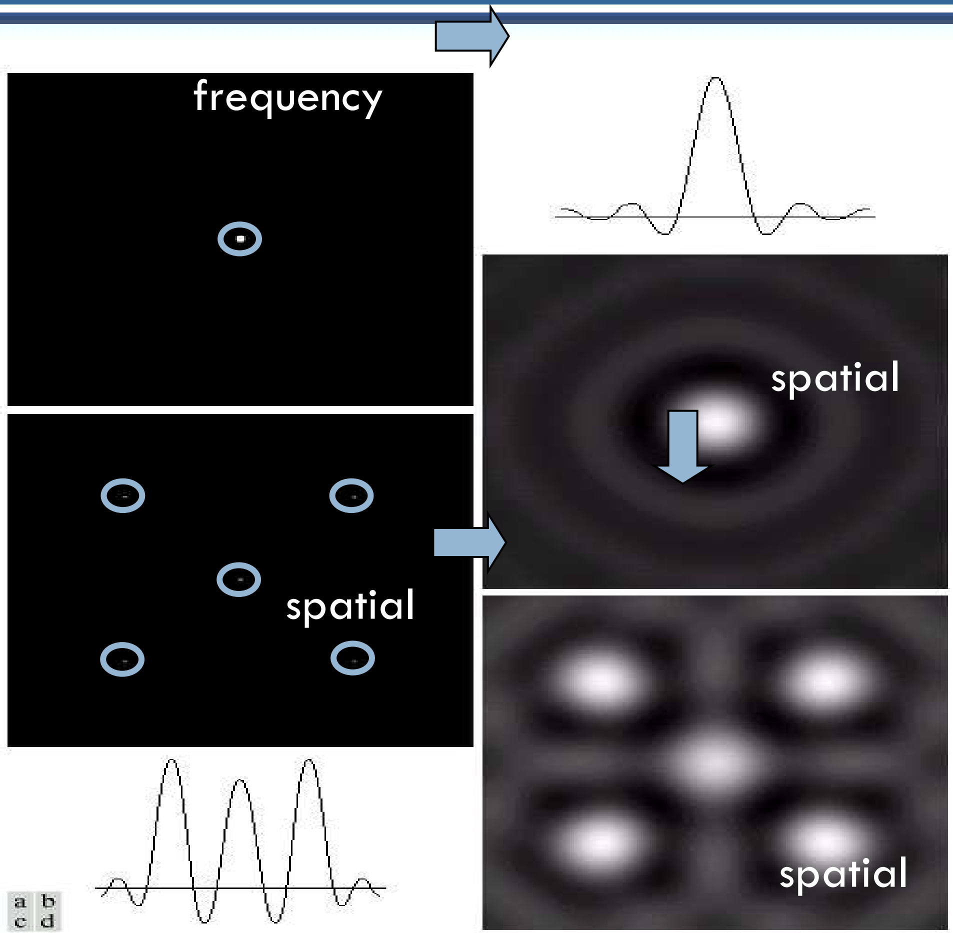
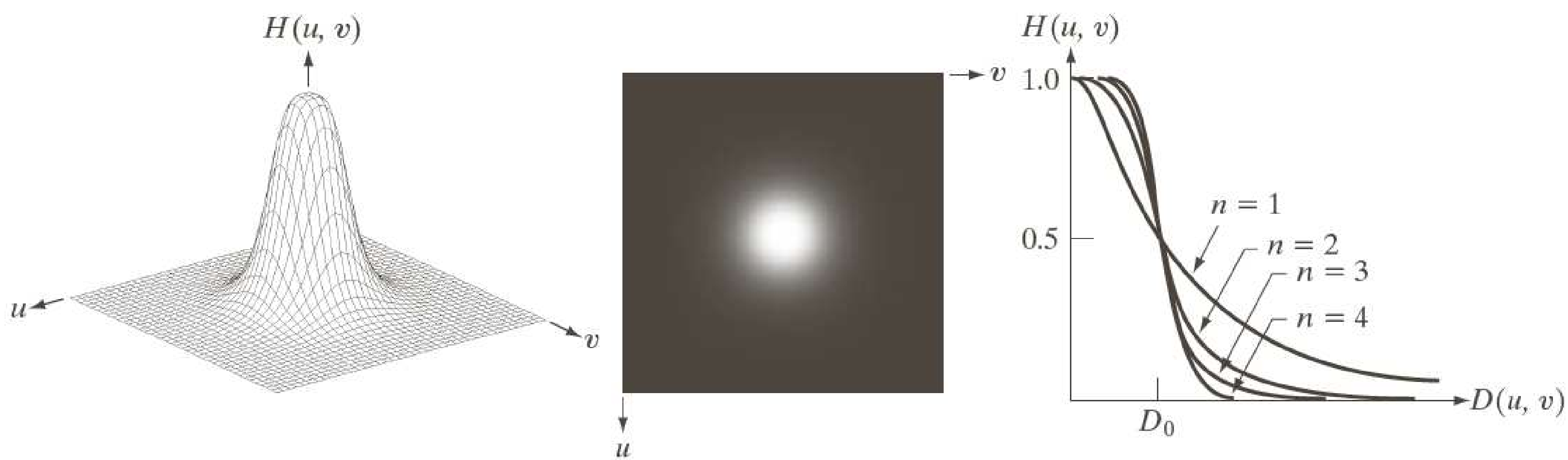


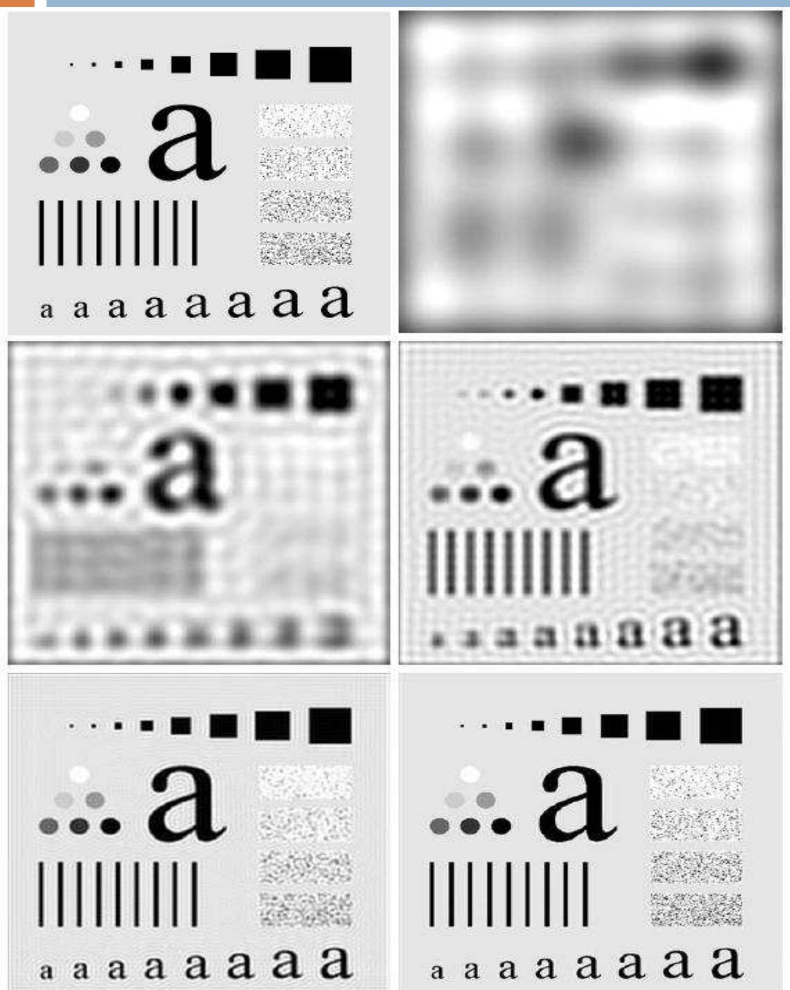
FIGURE 4.13 (a) A frequency-domain ILPF of radius 5. (b) Corresponding spatial filter (note the ringing). (c) Five impulses in the spatial domain, simulating the values of five pixels. (d) Convolution of (b) and (c) in the spatial domain.

Image Smoothing Using Filter Domain Filters: BLPF



a b c

FIGURE 4.44 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections of orders 1 through 4.



a b
c d
e f

FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.

a b
c d
e f

FIGURE 4.45 (a) Original image. (b)–(f) Results of filtering using BLPFs of order 2, with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Fig. 4.42.

Butterworth Lowpass Filters (BLPFs) Spatial Representation

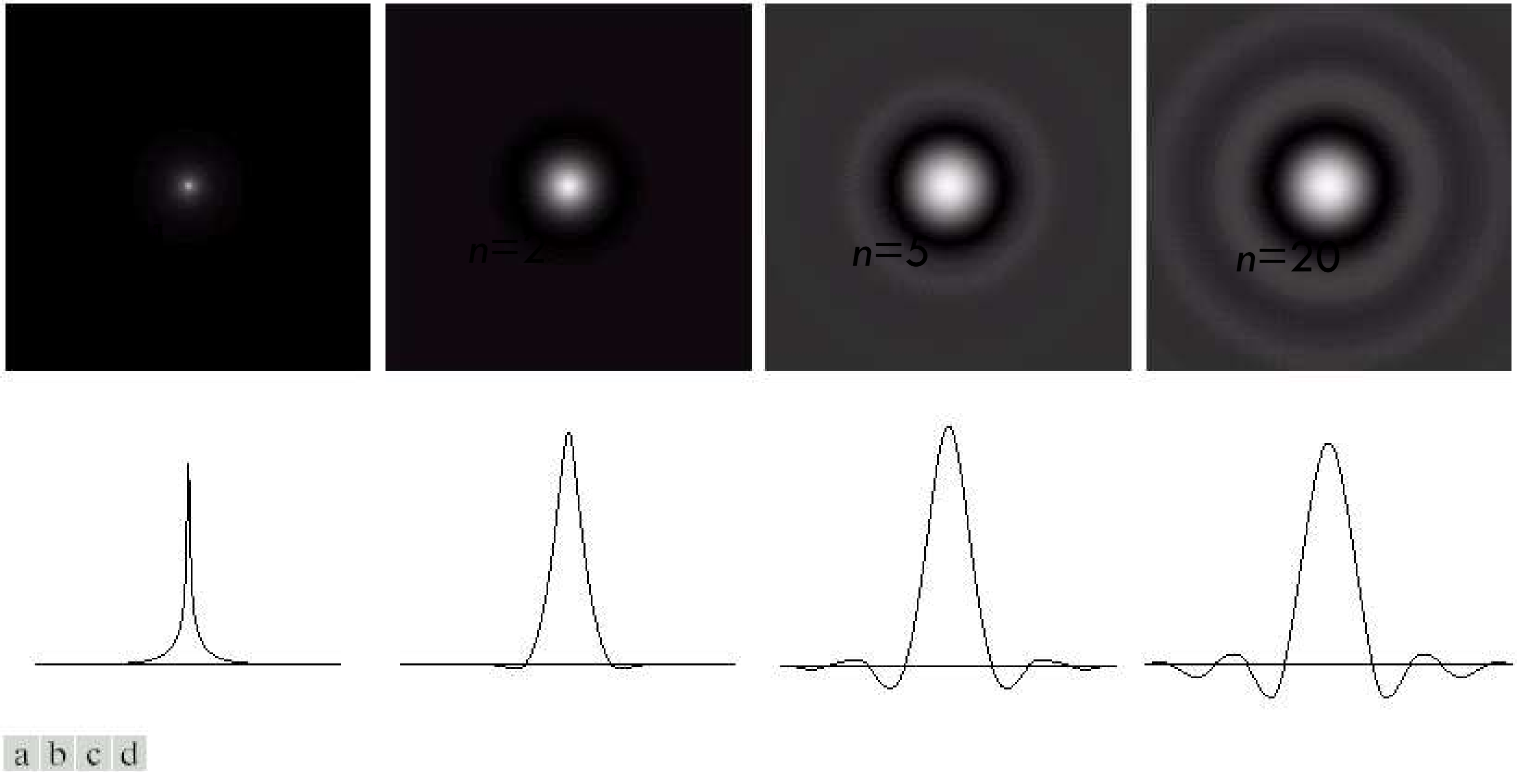
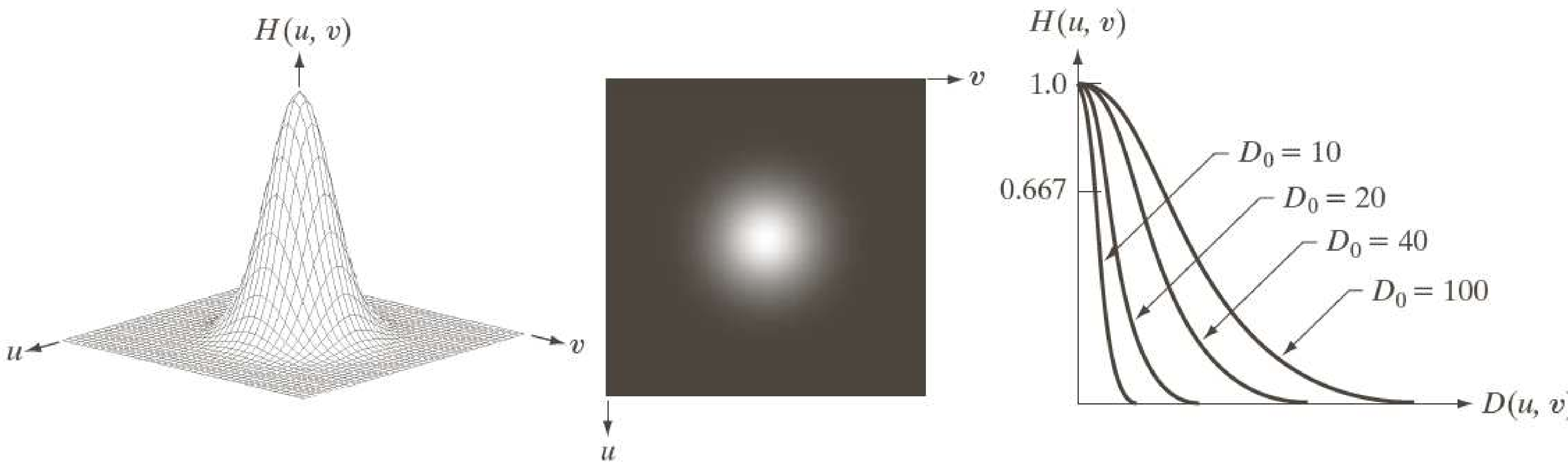


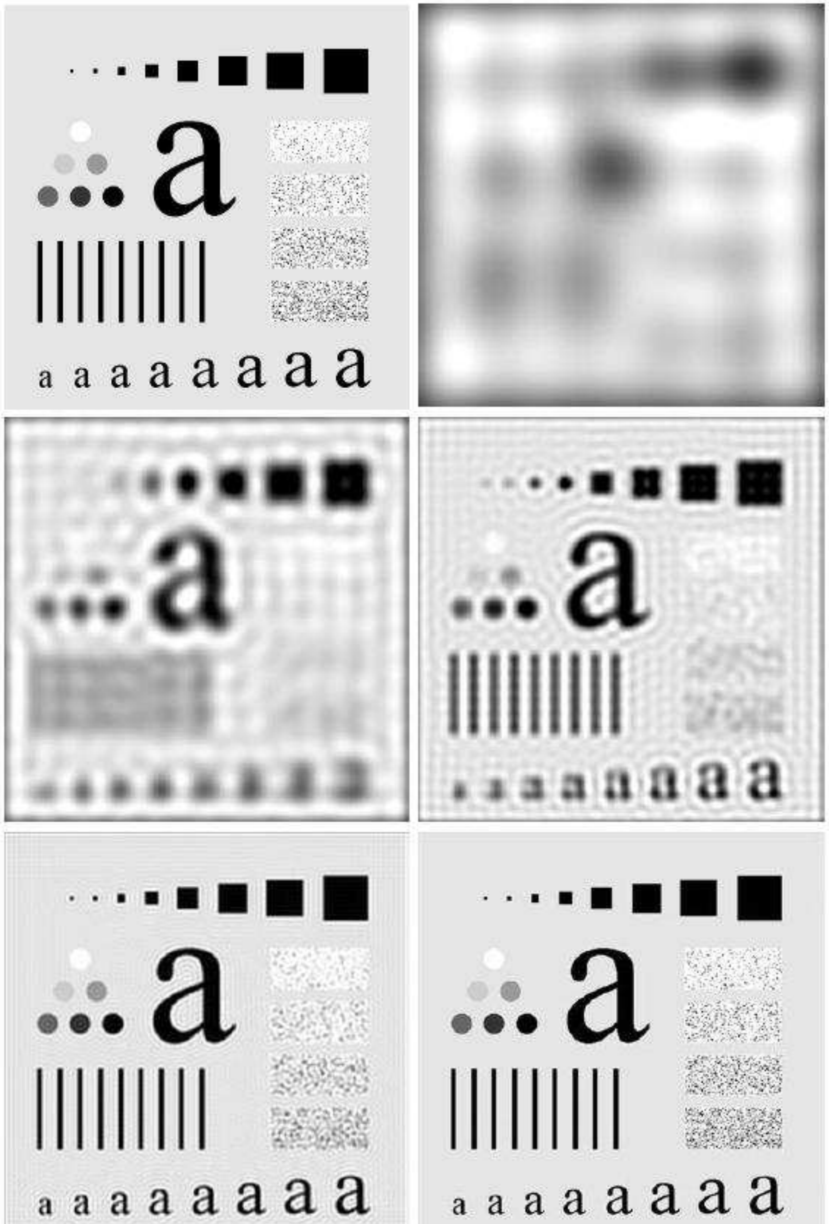
FIGURE 4.16 (a)–(d) Spatial representation of BLPFs of order 1, 2, 5, and 20, and corresponding gray-level profiles through the center of the filters (all filters have a cutoff frequency of 5). Note that ringing increases as a function of filter order.

Image Smoothing Using Filter Domain Filters: GLPF



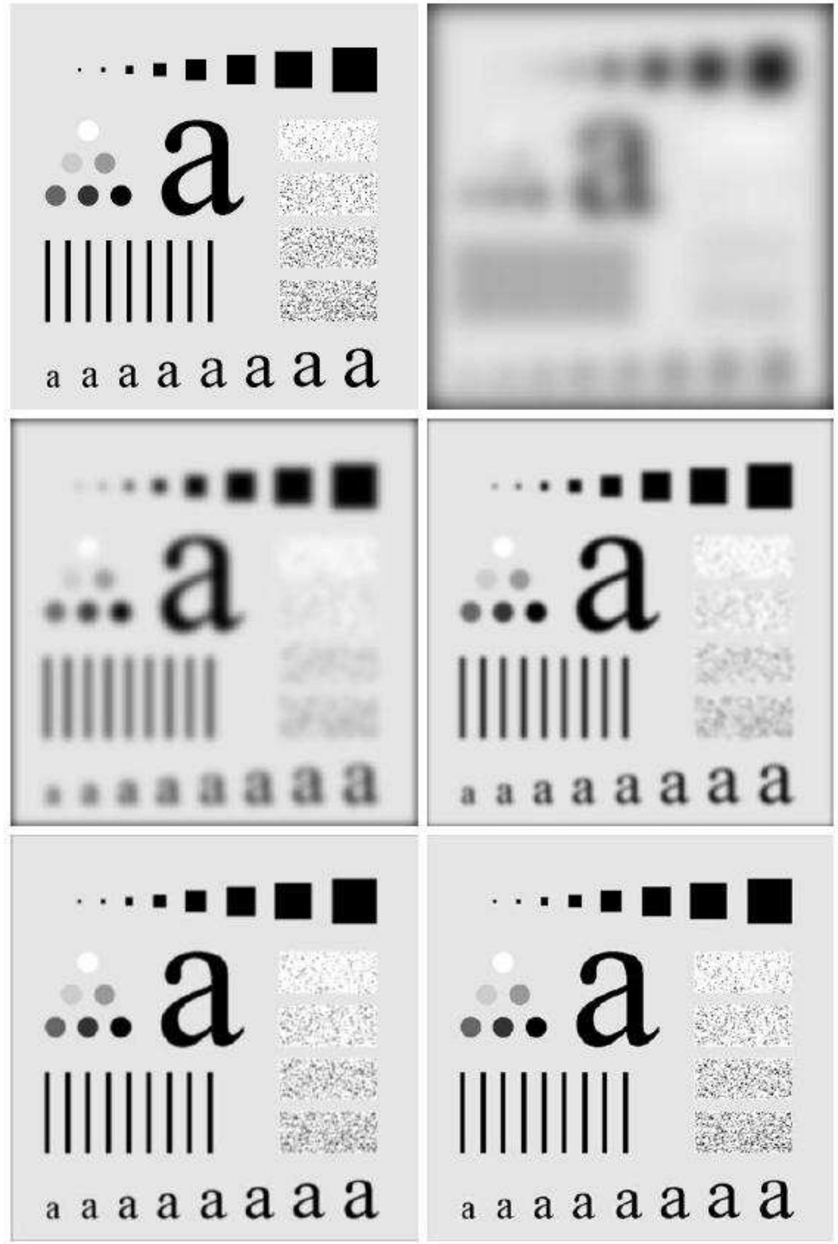
a b c

FIGURE 4.47 (a) Perspective plot of a GLPF transfer function. (b) Filter displayed as an image. (c) Filter radial cross sections for various values of D_0.



a b
c d
e f

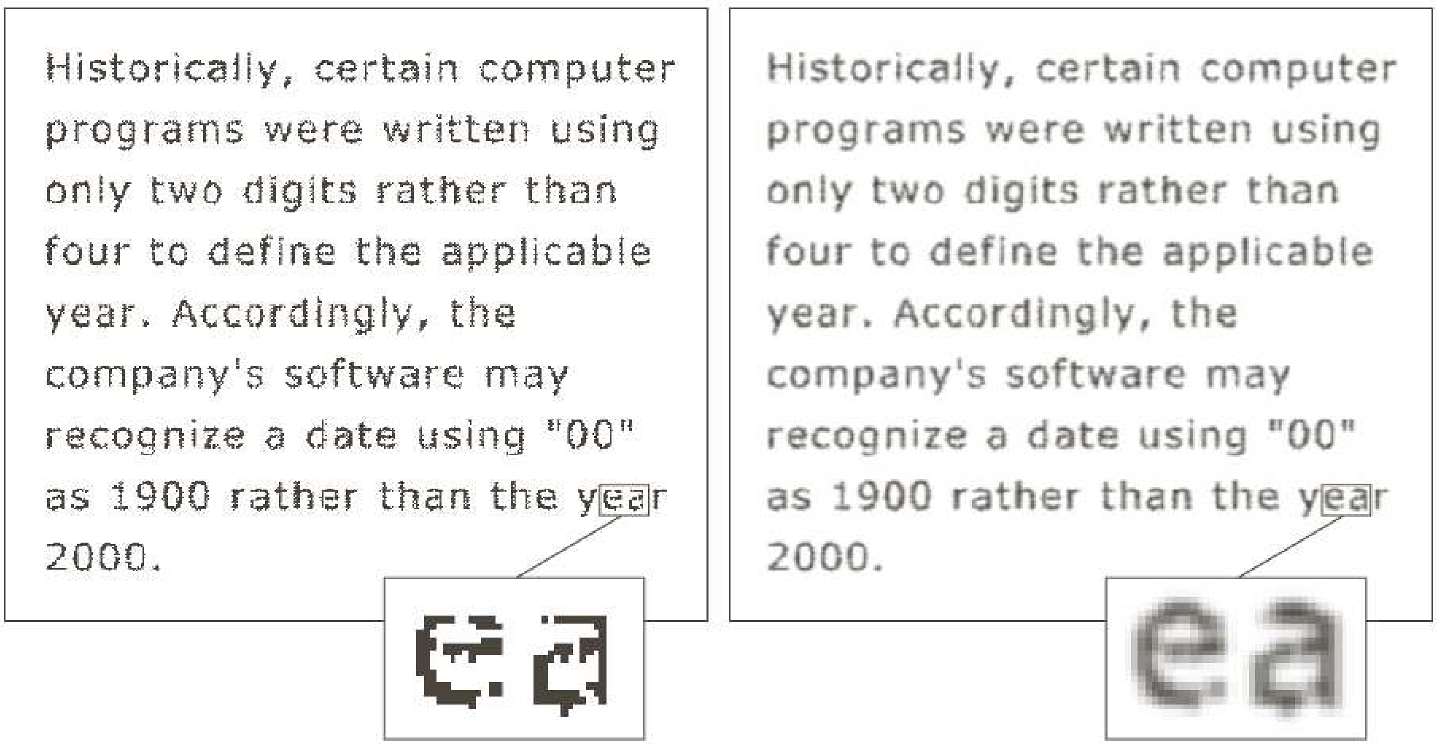
FIGURE 4.42 (a) Original image. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.41(b). The power removed by these filters was 13, 6.9, 4.3, 2.2, and 0.8% of the total, respectively.



a b
c d
e f

FIGURE 4.48 (a) Original image. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.41. Compare with Figs. 4.42 and 4.45.

Examples of smoothing by GLPF (1)



a b

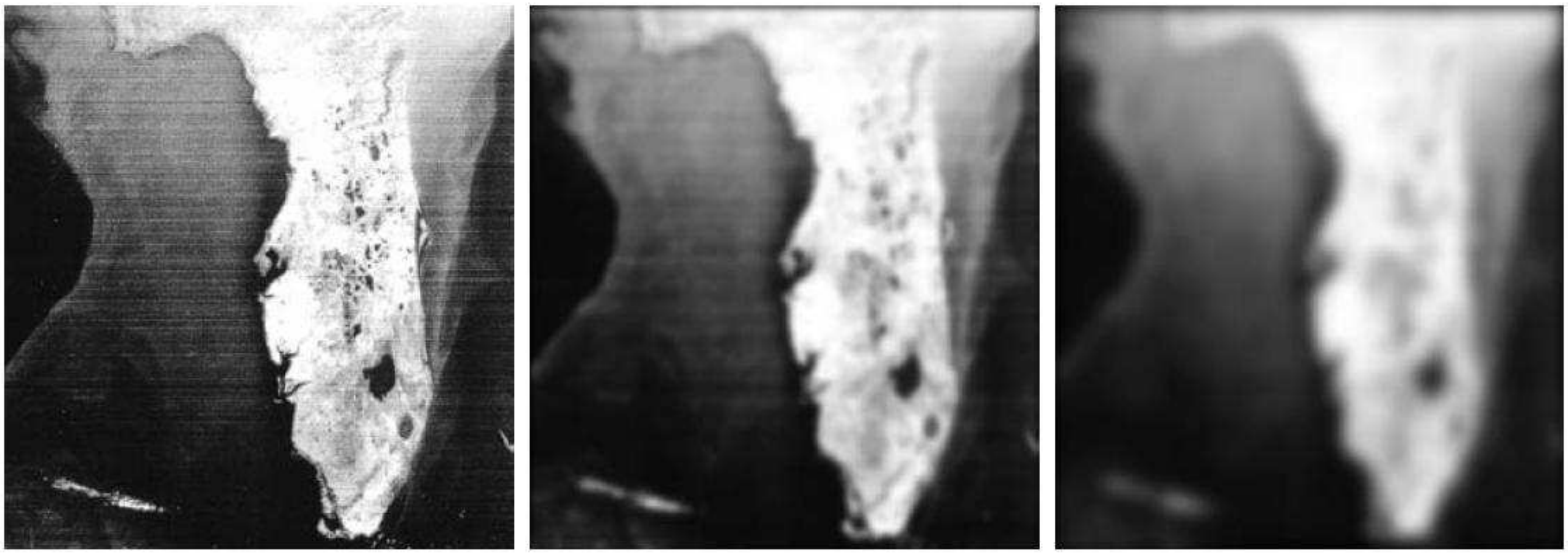
FIGURE 4.49
(a) Sample text of low resolution (note broken characters in magnified view). (b) Result of filtering with a GLPF (broken character segments were joined).

Examples of smoothing by GLPF (2)

a b c

FIGURE 4.50 (a) Original image (784 × 732 pixels). (b) Result of filtering using a GLPF with $D_0 = 100$. (c) Result of filtering using a GLPF with $D_0 = 80$. Note the reduction in fine skin lines in the magnified sections in (b) and (c).

Examples of smoothing by GLPF (3)



a b c

FIGURE 4.51 (a) Image showing prominent horizontal scan lines. (b) Result of filtering using a GLPF with $D_0 = 50$. (c) Result of using a GLPF with $D_0 = 20$. (Original image courtesy of NOAA.)

4.Image Sharpening Using Frequency Domain Filters

A high pass filter is obtained from a given low pass filter using

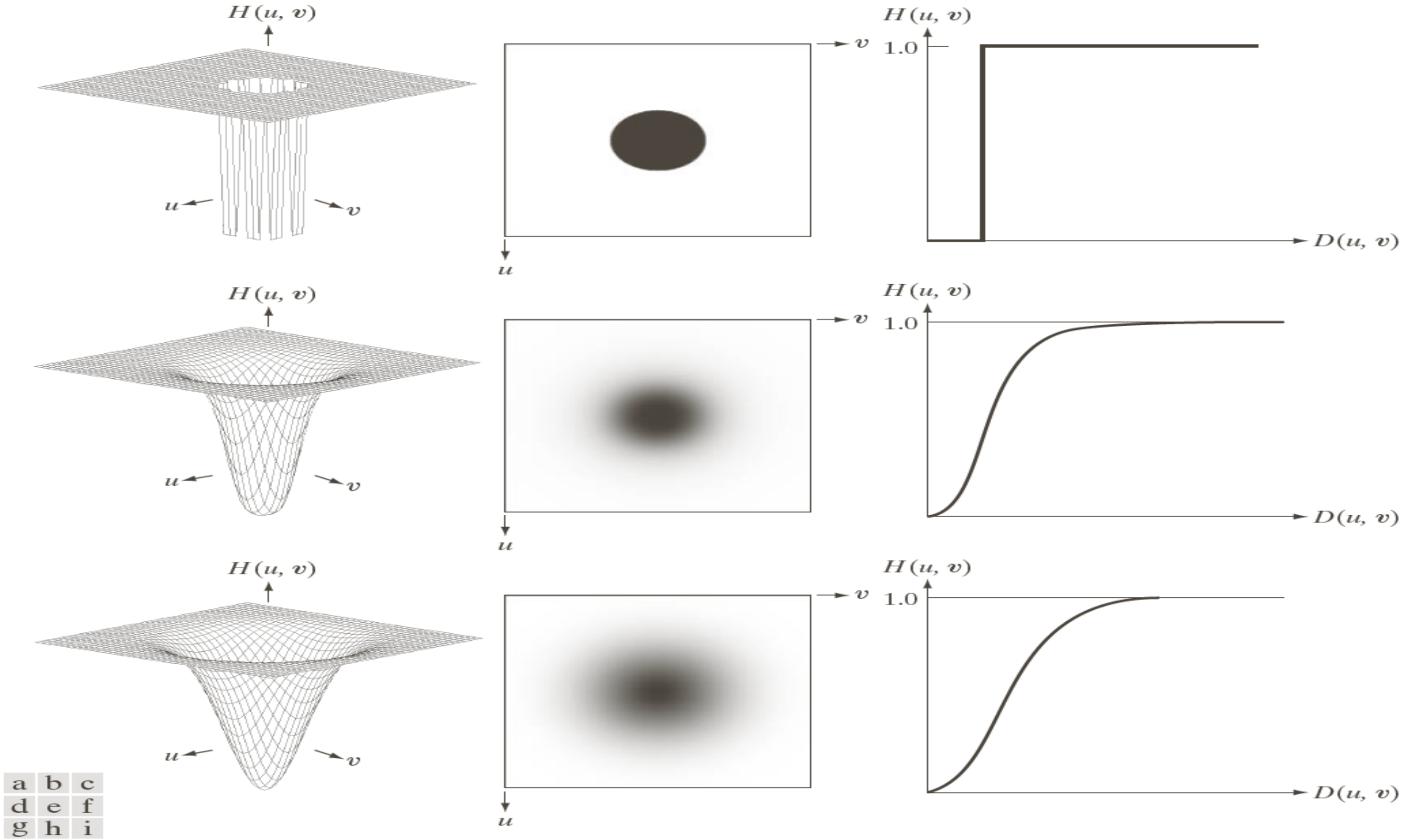


FIGURE 4.52 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

Sharpening Frequency Domain Filter

Ideal highpass filter

Butterworth highpass filter

Gaussian highpass filter

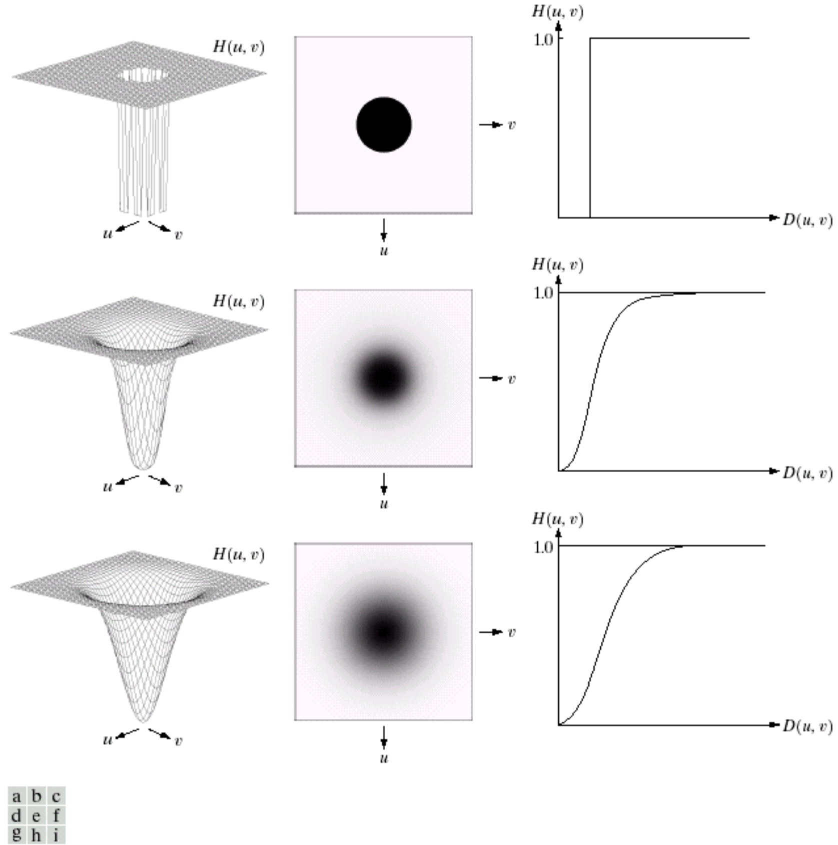


FIGURE 4.22 Top row: Perspective plot, image representation, and cross section of a typical ideal highpass filter. Middle and bottom rows: The same sequence for typical Butterworth and Gaussian highpass filters.

The Spatial Representation of Highpass Filters

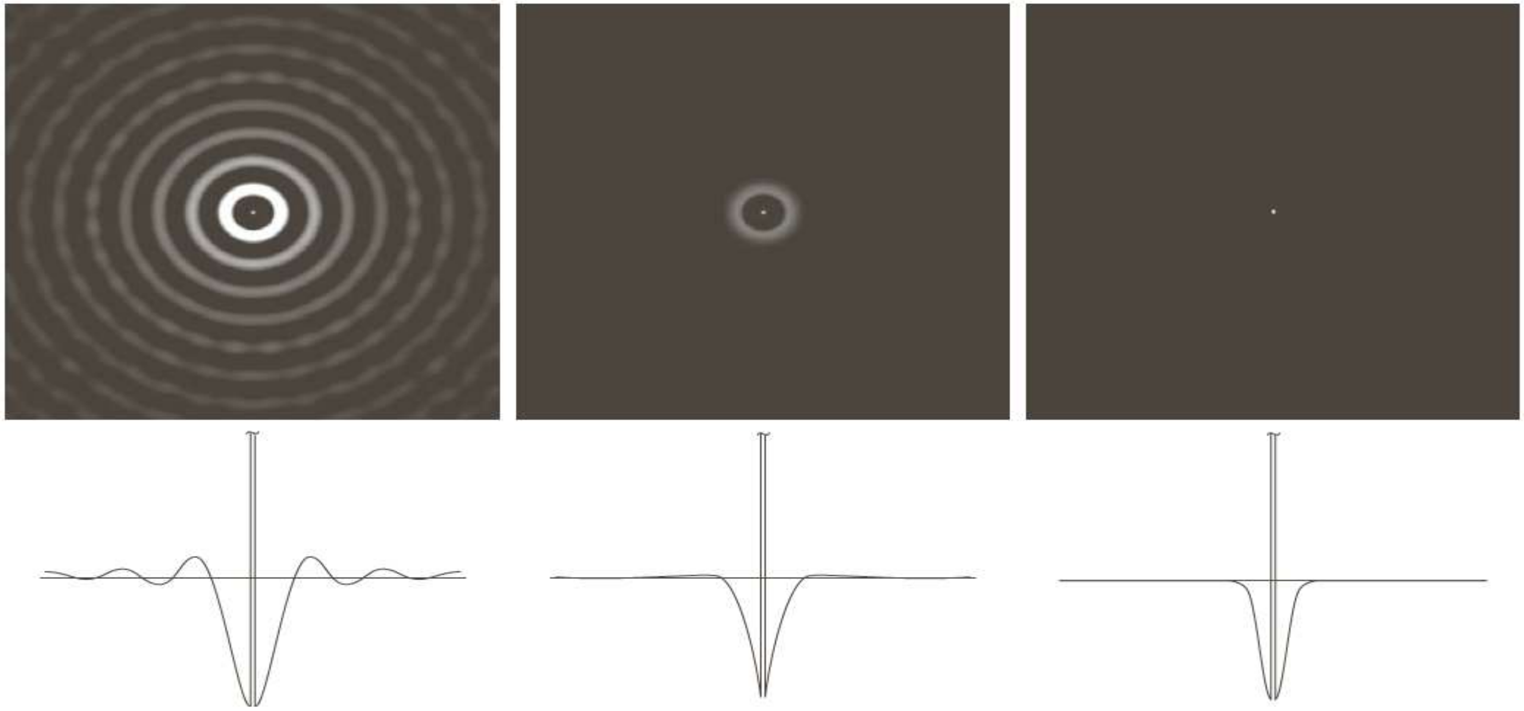


FIGURE 4.53 Spatial representation of typical (a) ideal, (b) Butterworth, and (c) Gaussian frequency domain highpass filters, and corresponding intensity profiles through their centers.

Filtering Results by IHPF

a b c

FIGURE 4.54 Results of highpass filtering the image in Fig. 4.41(a) using an IHPF with $D_0 = 30$, 60, and 160.

Filtering Results by BHPF

a b c

FIGURE 4.55 Results of highpass filtering the image in Fig. 4.41(a) using a BHPF of order 2 with $D_0 = 30, 60,$ and 160, corresponding to the circles in Fig. 4.41(b). These results are much smoother than those obtained with an IHPF.

Filtering Results by GHPF

a b c

FIGURE 4.56 Results of highpass filtering the image in Fig. 4.41(a) using a GHPF with $D_0 = 30, 60$, and 160, corresponding to the circles in Fig. 4.41(b). Compare with Figs. 4.54 and 4.55.

5. Homomorphic Filtering

An image consists of two components:

(1) illumination, that is how much light is shined unto the object,

$$0 < i(x, y) < \infty,$$

and is independent of the object itself;

(2) reflectance, that is the fraction of light that is reflected by the object,

$$0 < r(x, y) < 1,$$

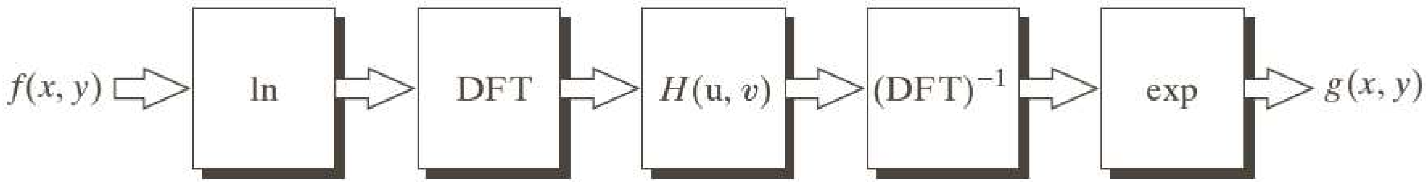
and is obviously a property of the object in question

when the object is black $r(x, y) = 0$

when the object is white $r(x, y) = 1$;

Homomorphic Filtering

FIGURE 4.60
Summary of steps in homomorphic filtering.



The illumination component of an image generally is characterized by slow spatial variations, while the reflectance component tends to vary abruptly

These characteristics lead to associating the low frequencies of the Fourier transform of the logarithm of an image with illumination the high frequencies with reflectance.

Homomorphic Filtering

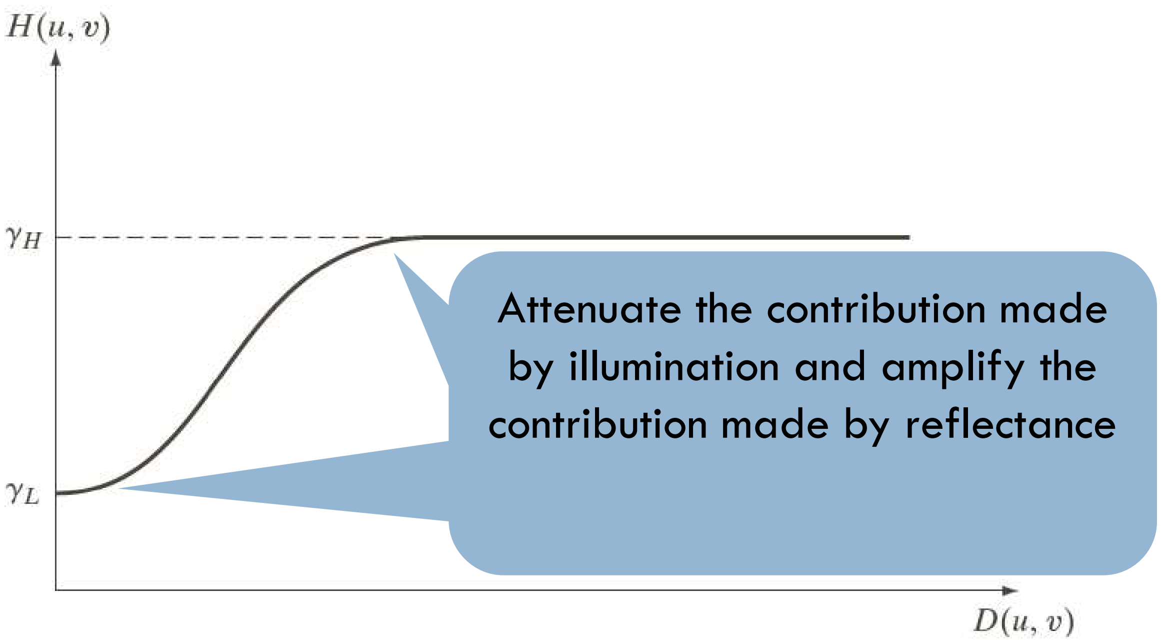
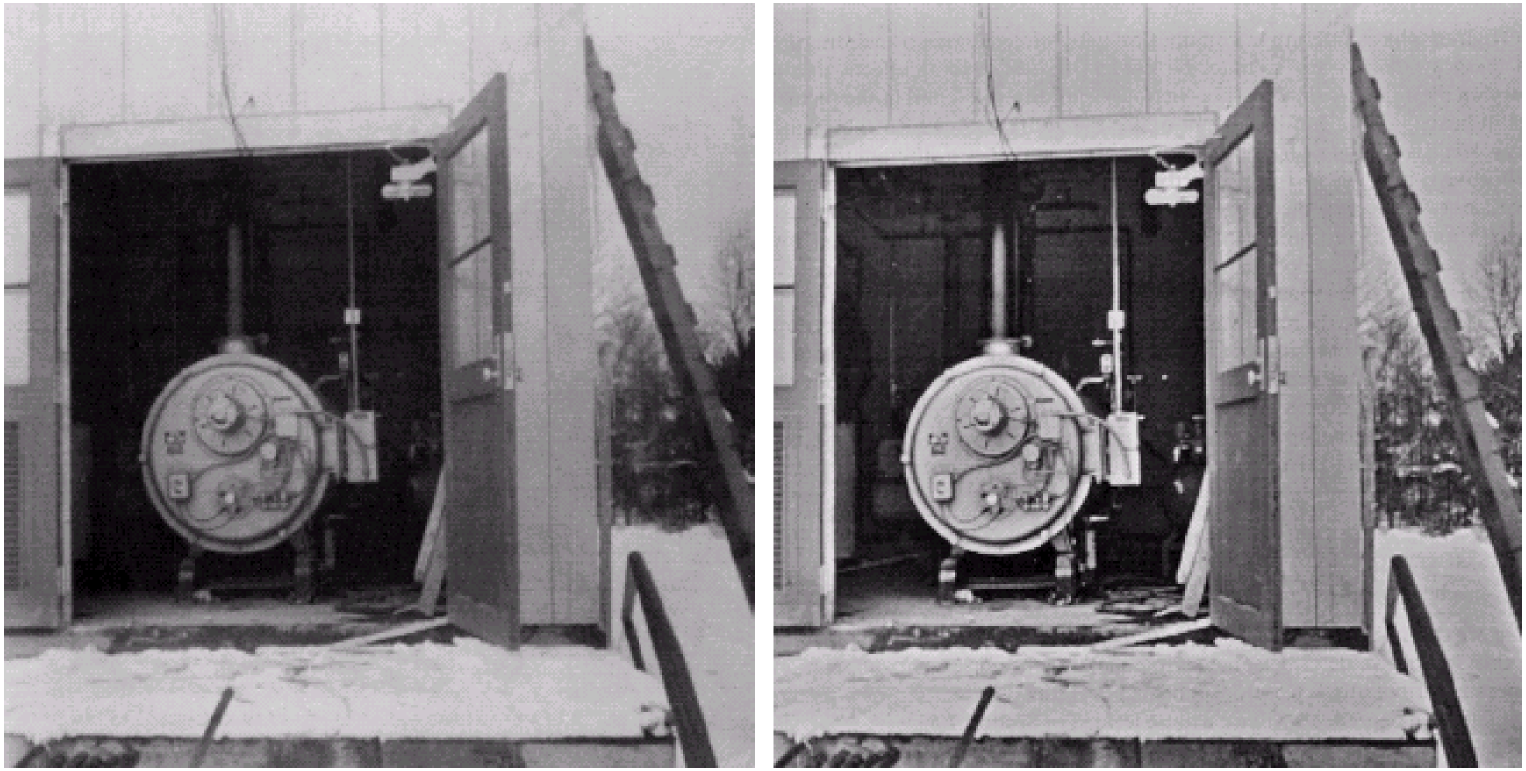


FIGURE 4.61 Radial cross section of a circularly symmetric homomorphic filter function. The vertical axis is at the center of the frequency rectangle and $D(u, v)$ is the distance from the center.

Homomorphic Filtering

a b

FIGURE
(a) Original image. (b) Image processed by homomorphic filtering (note details inside shelter). (Stockham.)



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